

Ageing • Models • Diagnostics • Pack Design • Electromobility • Stationary Energy Storage • Energy System Analysis

Grundgebiete der Elektrotechnik 1 – Einführung in die Schaltungstechnik

Kapitel 1 – Was der Lehrstuhl für Elektrochemische Energiewandlung & Speichersystemtechnik macht

G1 1 V8

Prof. Dirk Uwe Sauer

Lehrstuhl für Elektrochemische Energiewandlung und Speichersystemtechnik

iSEA Stromrichtertechnik und Elektrische Antriebe | RWTH AACHEN UNIVERSITY

Was der Lehrstuhl für

Elektrochemische Energiewandlung und Speichersystemtechnik

(Prof. Dr. Dirk Uwe Sauer)

macht

Lehrstuhl als Teil des Instituts für Stromrichtertechnik und Elektrische Antriebe (ISEA)



- Univ.-Prof. Dr.-Ing. Christoph van der Broeck
  - Chair for Power Electronic Systems
  - Director of ISEA

since 2026



- Univ.-Prof. Dr. ir. Dr. h. c. Rik De Doncker
  - Chair for Power Electronics and Electrical Drives
  - Director Research Campus “Flexible Electrical Networks”

since 1996



- Univ.-Prof. Dr. rer. nat. Dirk Uwe Sauer
  - Chair for Electrochemical Energy Conversion and Storage Systems
  - Director Research Center CARL

since 2003



- Univ.-Prof. Dr. rer. nat. Egbert Figgemeier
  - Teaching and Research Area for Ageing and Lifetime Prediction of Batteries
  - Research Group Leader “Cell Systems and Ageing” @ Helmholtz Center Jülich

since 2016

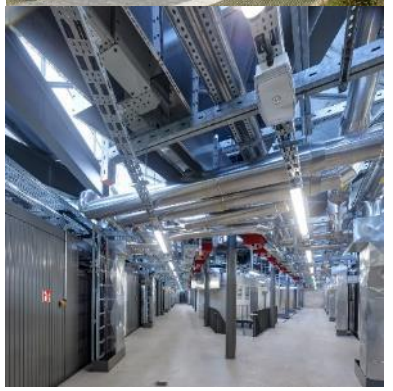


Jun.-Prof. Dr.-Ing. Weihan Li
  - Artificial Intelligence and Digitalization for Batteries

since 2025







Aufgaben eines Lehrstuhls

- **Lehre**
  - Vorlesungen & Übungen
  - Praktika & Seminare
  - Betreuung von Bachelor- und Masterarbeiten
  - Ausbildung von Wissenschaftlerinnen und Wissenschaftlern (Promotionen)
- **Berufliche Weiterbildung**
  - Seminarangebote für Unternehmen zur Weiterbildung der Mitarbeiterinnen und Mitarbeiter
- **Veranstaltung von Tagungen und Konferenzen**
  - Wissenschaftliche Konferenzen
  - Tagungen zum Netzwerken zwischen akademischen Einrichtungen und Industrie
- **Veranstaltungen für die Nachwuchsgewinnung**
  - Sommerschulen
  - Berufsinformationstage
  - „Girls Day“
- **Beteiligung an der universitären Selbstverwaltung**

Aufgaben eines Lehrstuhls

- **Forschung & Entwicklung**
  - Grundlagenforschung (primär für den Erkenntnisgewinn, in den Ingenieurswissenschaften aber eher selten)
  - Schreiben von wissenschaftlichen Veröffentlichungen, die den Stand der Technik dann für alle erweitern
  - Angewandte Forschung (Erforschung neuer Verfahren, Materialien, Algorithmen oder Software als Grundlage für neue Produkte oder Dienstleistungen)
  - Entwicklungsunterstützung für Unternehmen (konkrete Beteiligung an der Entwicklung neuer Produkte oder Dienstleistungen)
  - Entwicklung von Patenten zur Sicherung geistigen Eigentums
- **Unterstützung und Motivation zur Gründung von Spin-off-Firmen**
  - Am Lehrstuhl entwickeltes Know-how wird zusammen mit qualifizierten Mitarbeiterinnen und Mitarbeitern ausgegründet und kommerziell angeboten
  - Lehrstuhl unterstützt zunächst mit Infrastruktur (Labore, Werkstätten, Verwaltung) Kontakten und Übergabe von entwickelten Produkten, Modellen oder Software als Basis für die Unternehmung
- **Information und Beratung von Politik und Gesellschaft**
  - Wissenschaft hat eine gesellschaftliche Verantwortung und wird zu wesentlichen Teilen aus Steuergeldern bezahlt

Center for Ageing, Reliability and Lifetime Prediction for Electrochemical and Power Electronic Systems – Supporting battery applications, production and material design



- State government and Federal government invest about 110 Million Euros for building this center in Aachen
- 5000 m^2 of net area for laboratories and offices
- 5000 battery testing channels and environmental stress lab
- Chemical labs for post-mortem analysis and material analysis
- Clean room & Dry room, CT & Microscopy
- Largest university research group in Europe for battery system technologies
  - More than 30 years of experience in battery system technologies
  - More than 80 researchers
  - More than 10 million € total budget per year


Prof. Martin Winter


Prof. Dirk Uwe Sauer


Prof. Egbert Figgemeier


Prof. Joachim Mayer


Prof. Hans-Georg Steinrück


Prof. Christoph Broeckmann


PD Dr. Gunther Brunklaus


Prof. Christoph van der Broeck


Prof. Maria Kateri


Prof. Weihan Li

- Interdisciplinary team from chemistry, physics, mathematics, material science, electrical and mechanical engineering

Center for Ageing, Reliability and Lifetime Prediction for Electrochemical and Power Electronic Systems – Supporting battery applications, production and material design

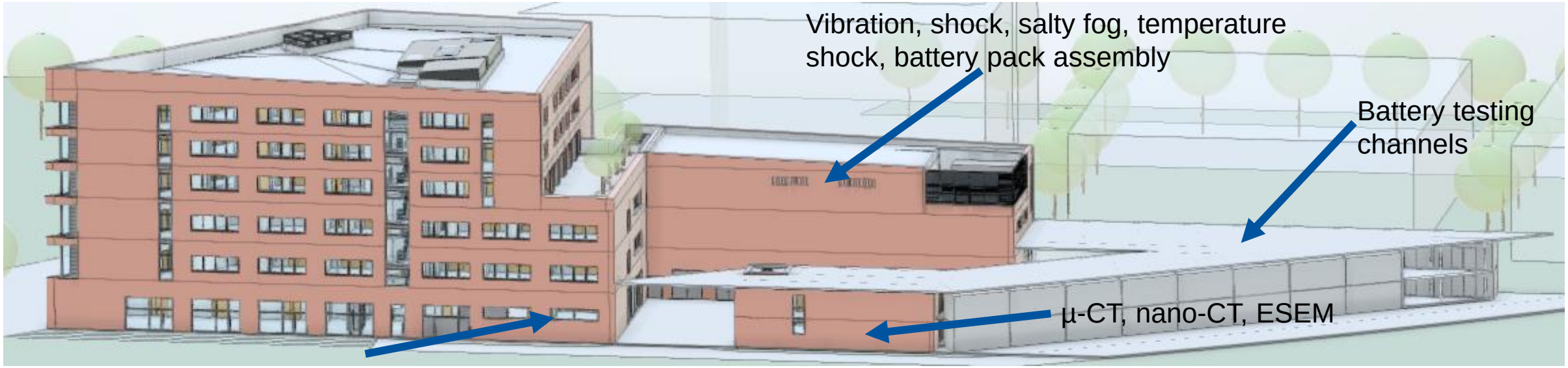


Center for Ageing, Reliability and Lifetime Prediction for Electrochemical and Electronics systems – Supporting battery applications and battery production

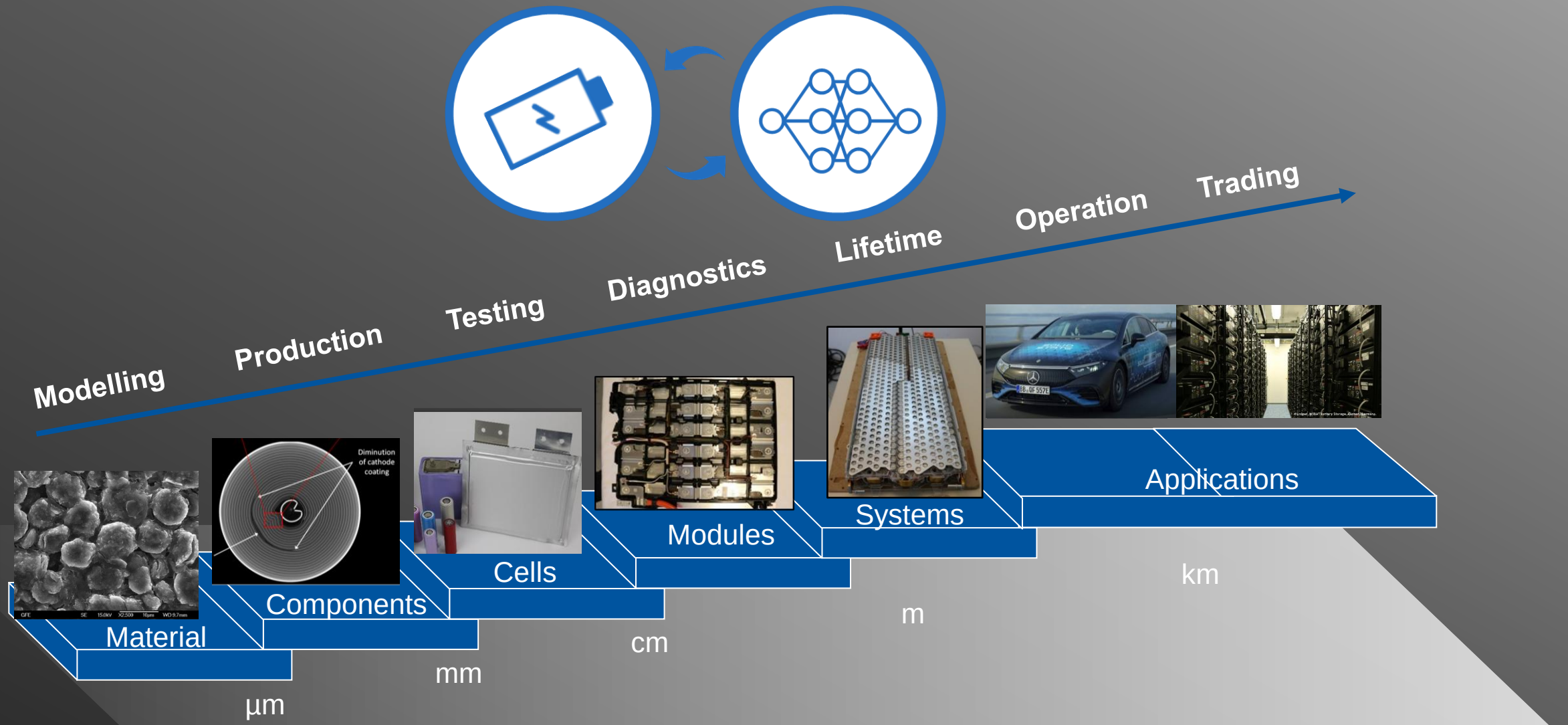
- 2150m² office area
- 2750m² laboratory area
- >150 researchers
- 110 Mio. € investment
- Opened in 2022



- >5000 battery testing channels
- Environmental stress lab
- Chemistry lab
- Clean room & Drying room
- CT & Microscopy

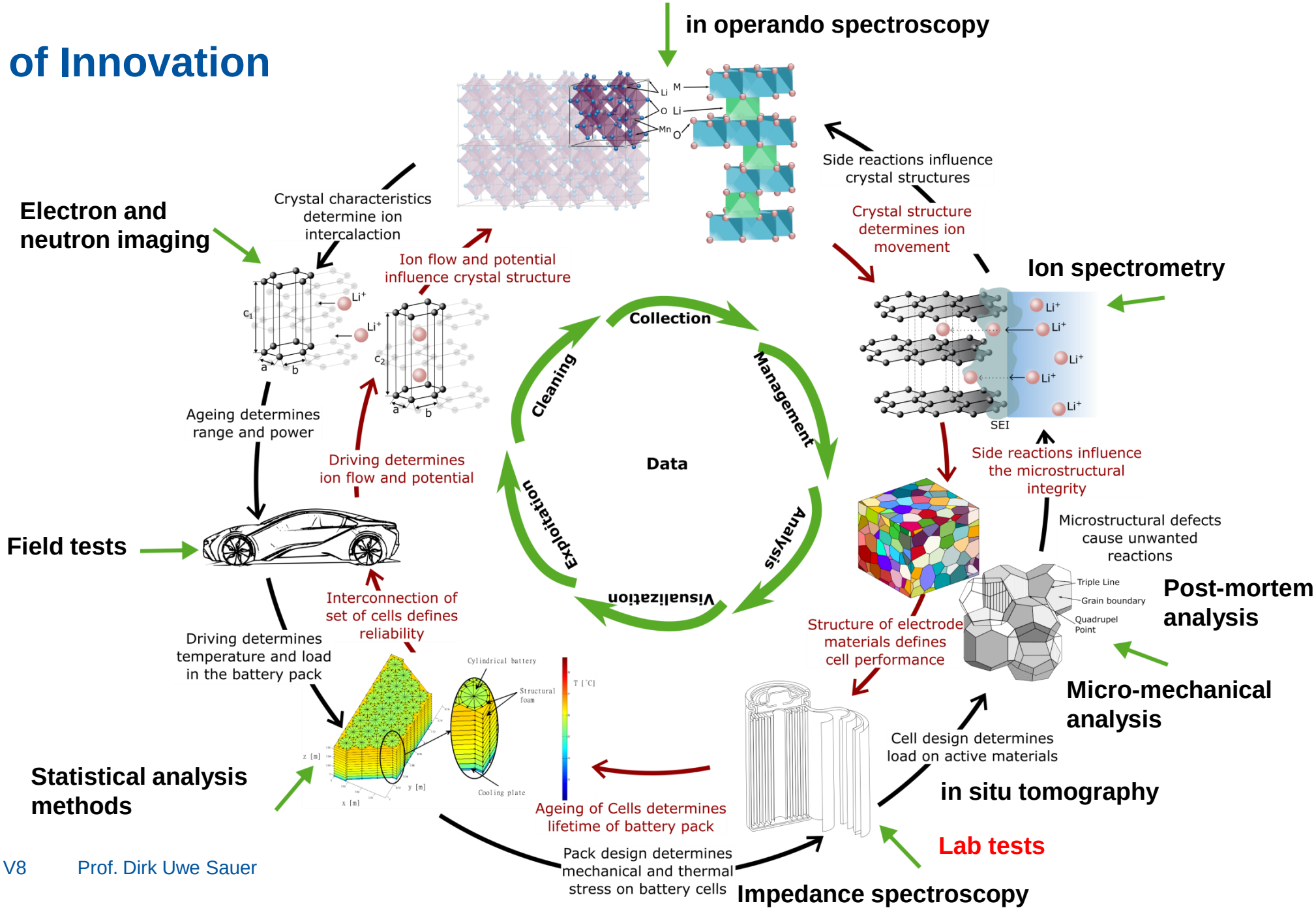


Battery research at CARL –
Multi disciplinary. Physics based. Experimental. Data. Artificial Intelligence



CARL Center for Ageing, Reliability and Lifetime Prediction of Electrochemical and Power Electronic Systems
RWTH AACHEN UNIVERSITY

Circle of Innovation



Battery Sections at ISEA



Ageing and Lifetime Prediction of Batteries

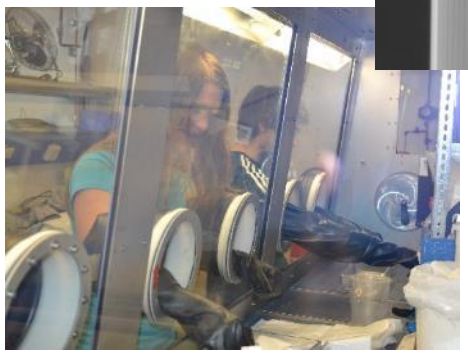
Electrochemical Energy Conversion and Storage Systems (ESS)

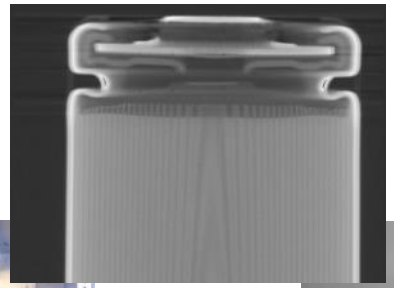


Artificial Intelligence and Digitalization for Batteries



Advanced electrode manufacturing	Ageing and Lifetime Prediction of Batteries	Battery Modelling, Analytics and Lifetime Prediction	Battery System Design and Vehicle Integration	Battery Grid Integration and Storage System Analysis	Artificial Intelligence for Batteries



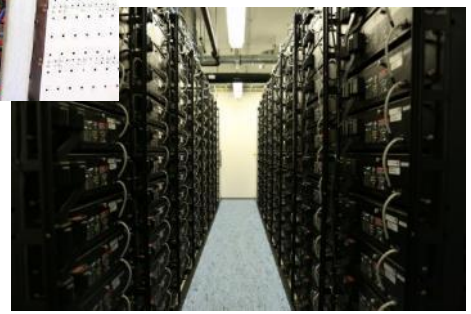




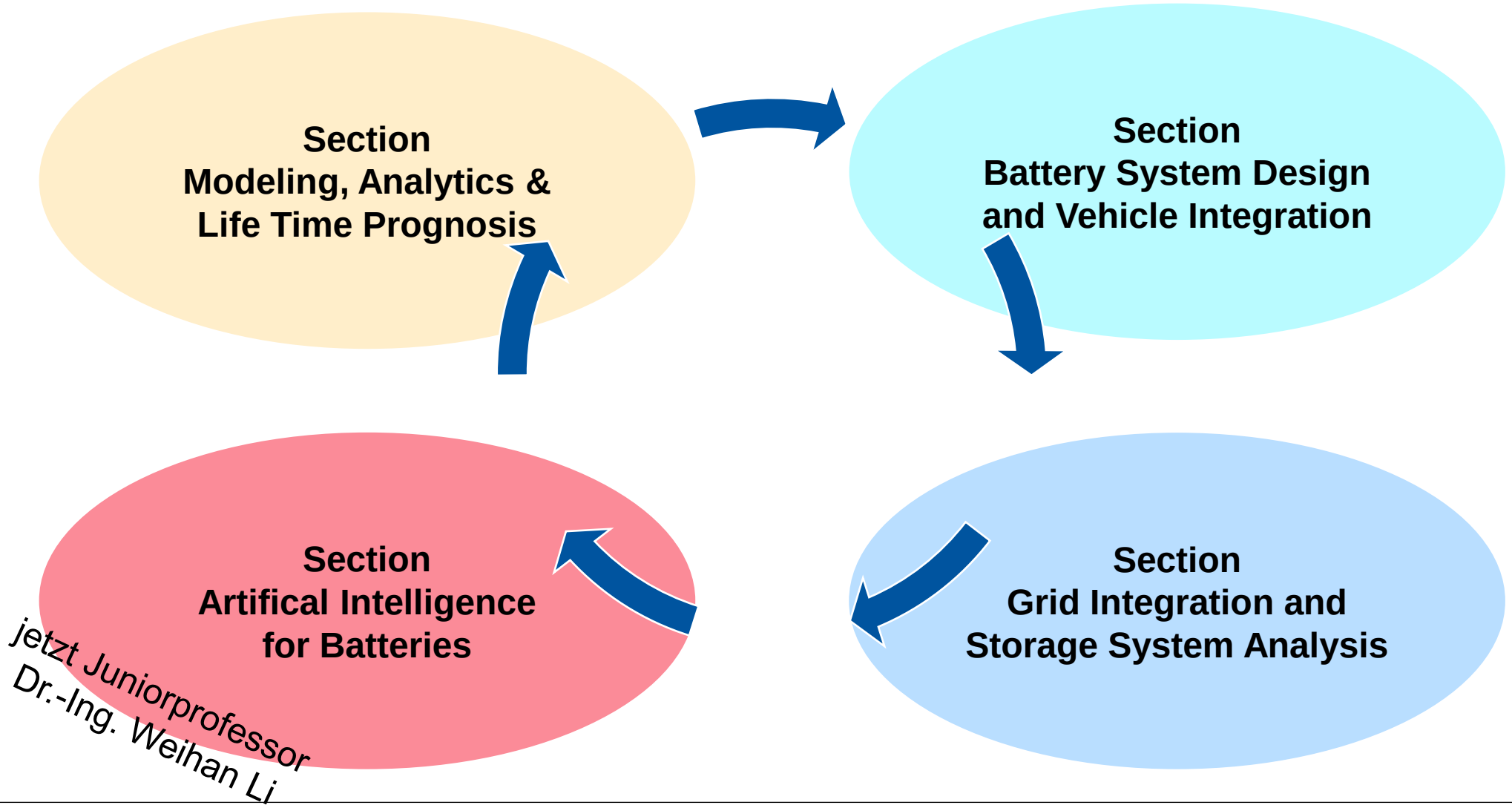




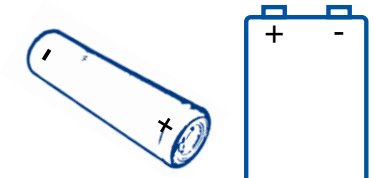




Chair Electrochemical Energy Conversion and Storage Systems – Prof. Dirk Uwe Sauer

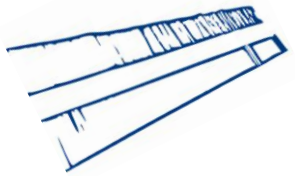


Overview: Battery Analytics in the Laboratory

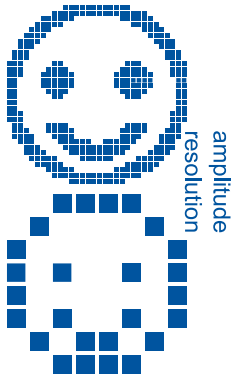


Open



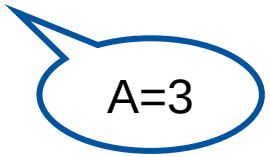








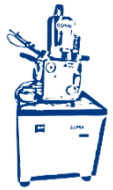






Analyse

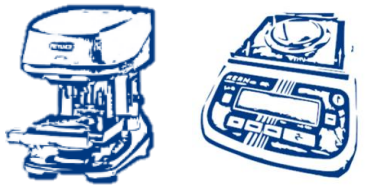
Evaluate



Material Composition



Lithium Distribution



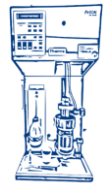
Surface Structure/Topology



Electrolyte Composition



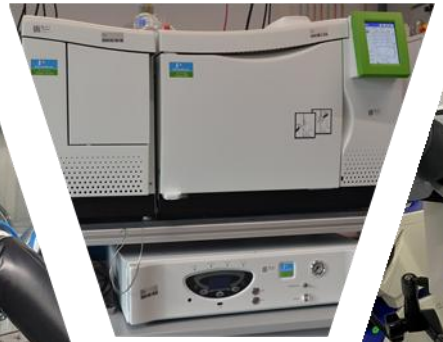
Geometrical Data

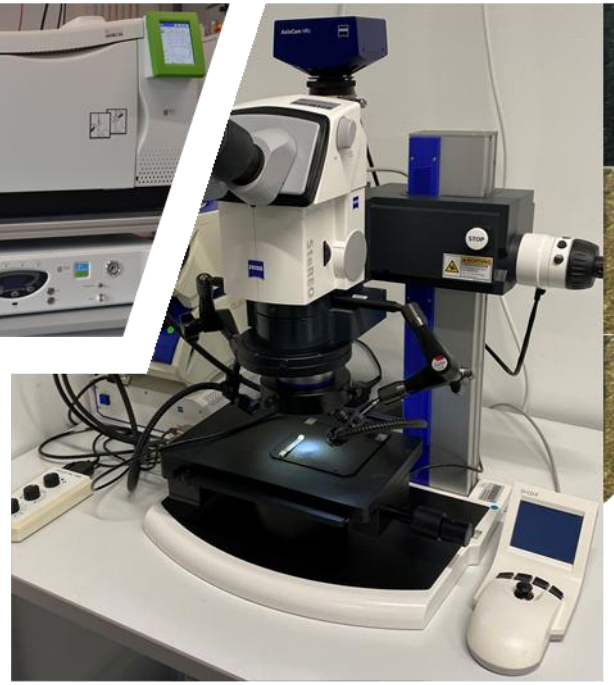


Porosity

Post Mortem Laboratory

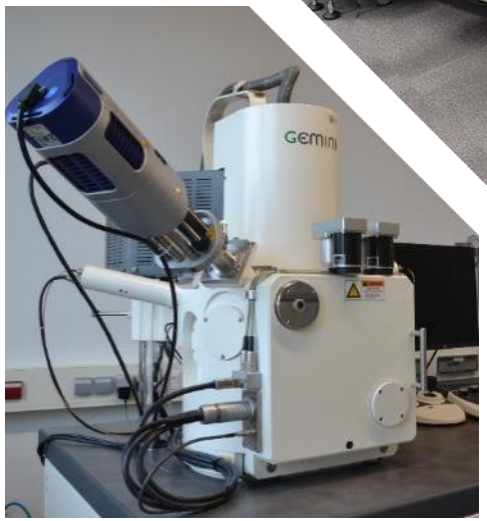




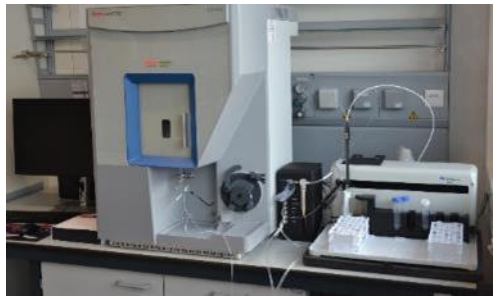












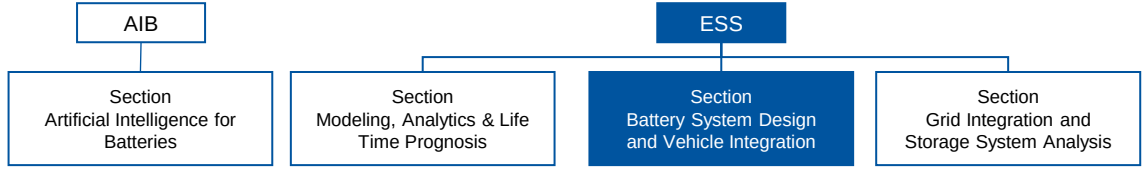


Overview Battery Testing



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Feedback made in CARL

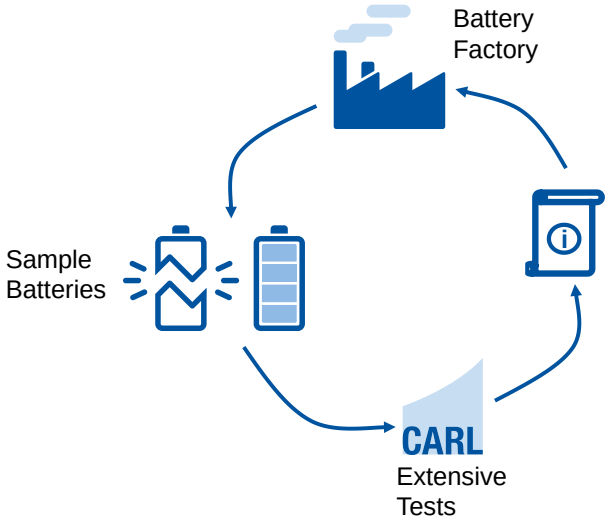


- Over 5 000 circuits
  - From nA to kA
  - From mV to kV
- Hundreds of climate chambers
- Circuits with EIS
- Temperature logging at each battery



- Post mortem analysis
- Mechanical stress tests

- New Parameters / Chemistries
- Ramp Up Production
- Quality Assurance

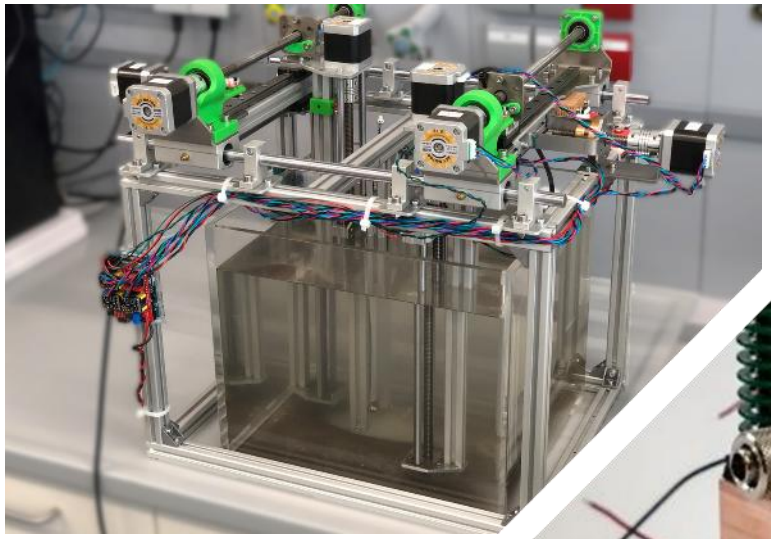


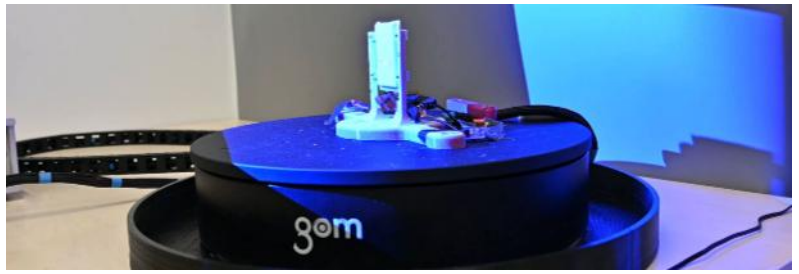
- Influence of ripple current on battery aging
  - Various materials
  - AC frequency
  - AC amplitude
- In-house designed ripple current generator

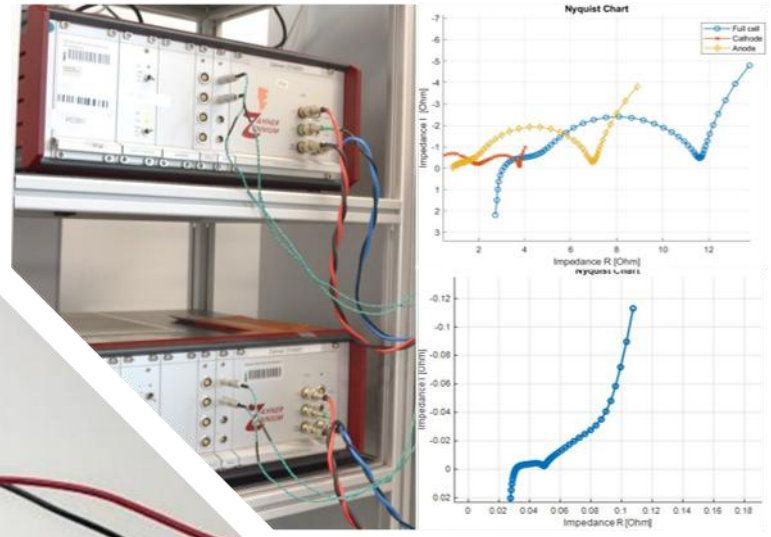




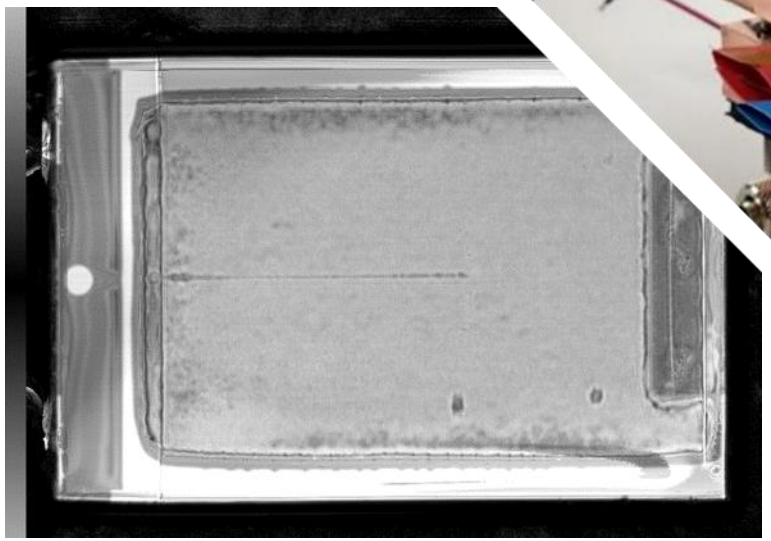
Overview Battery Sensors

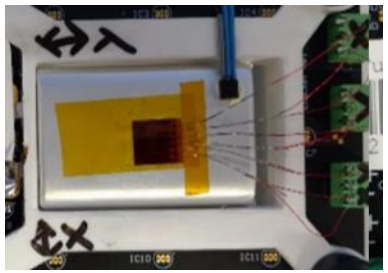


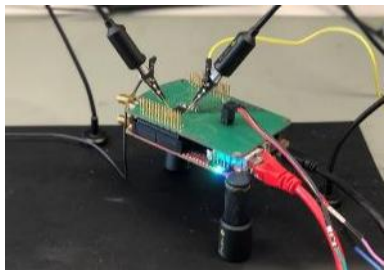




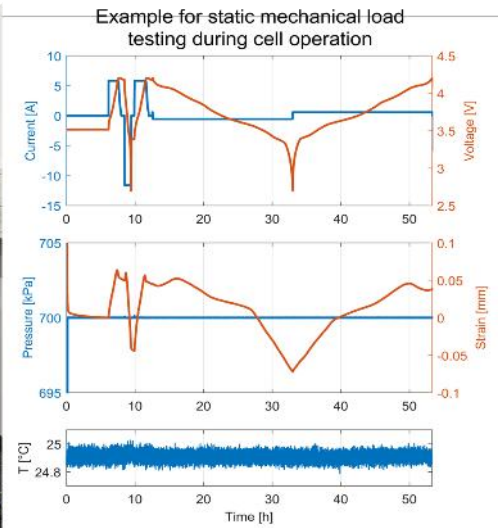






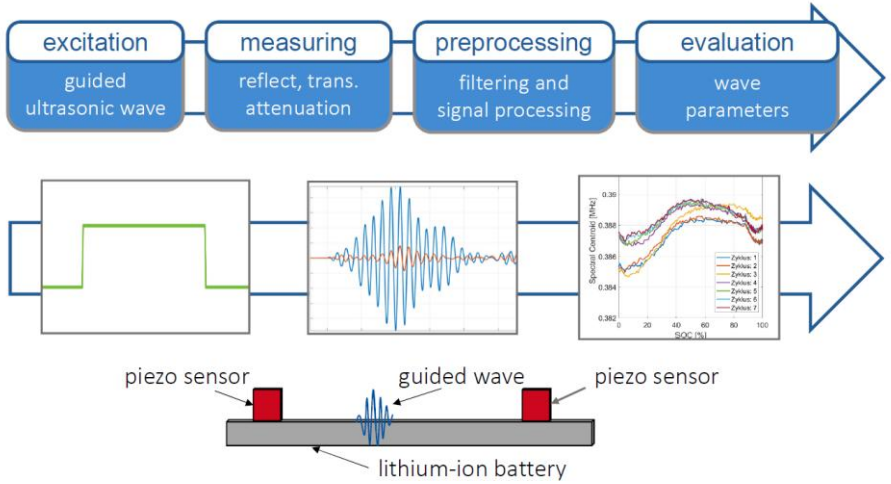




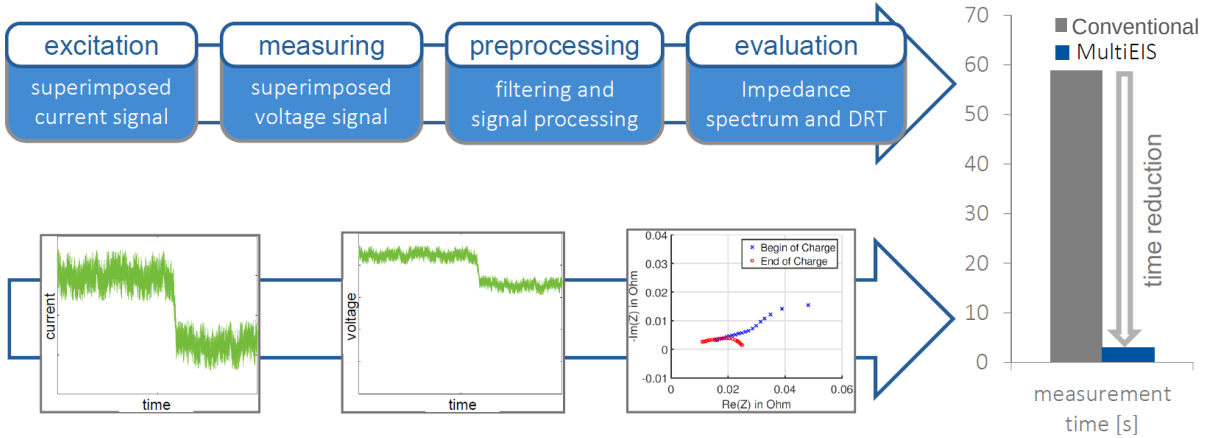


Battery Sensors - Overview

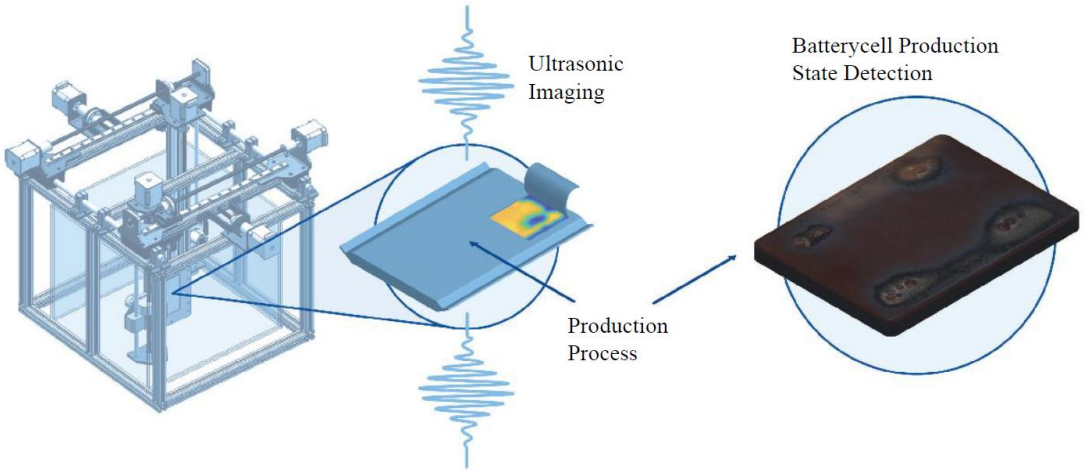
Ultra sound Sensing in Operando



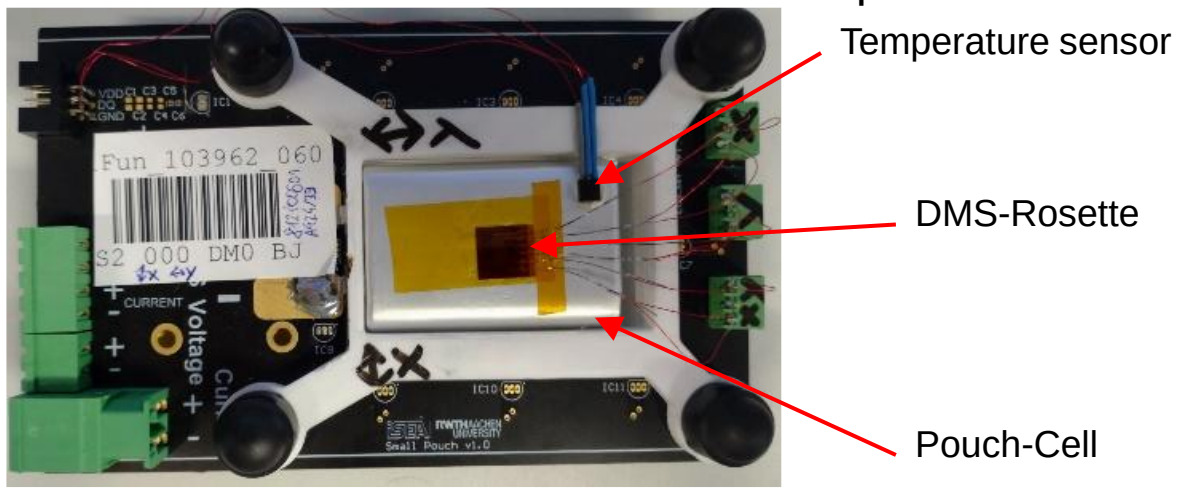
Multi-Sine EIS



Ultra sound Imaging



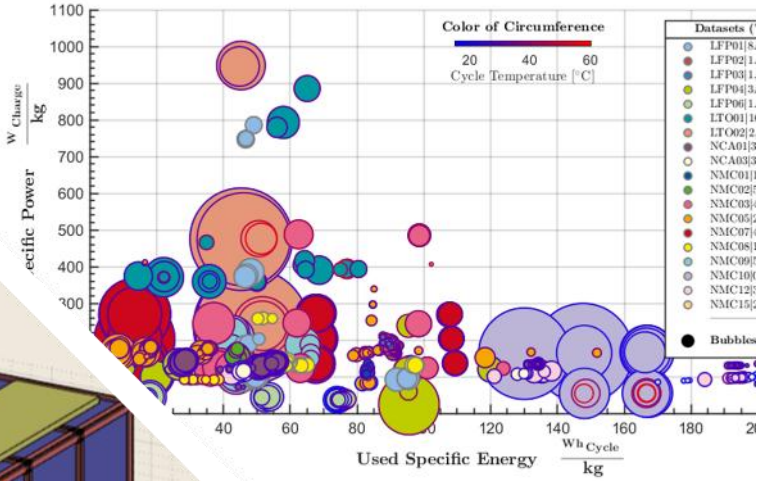
Dilatation Measurement Stripes

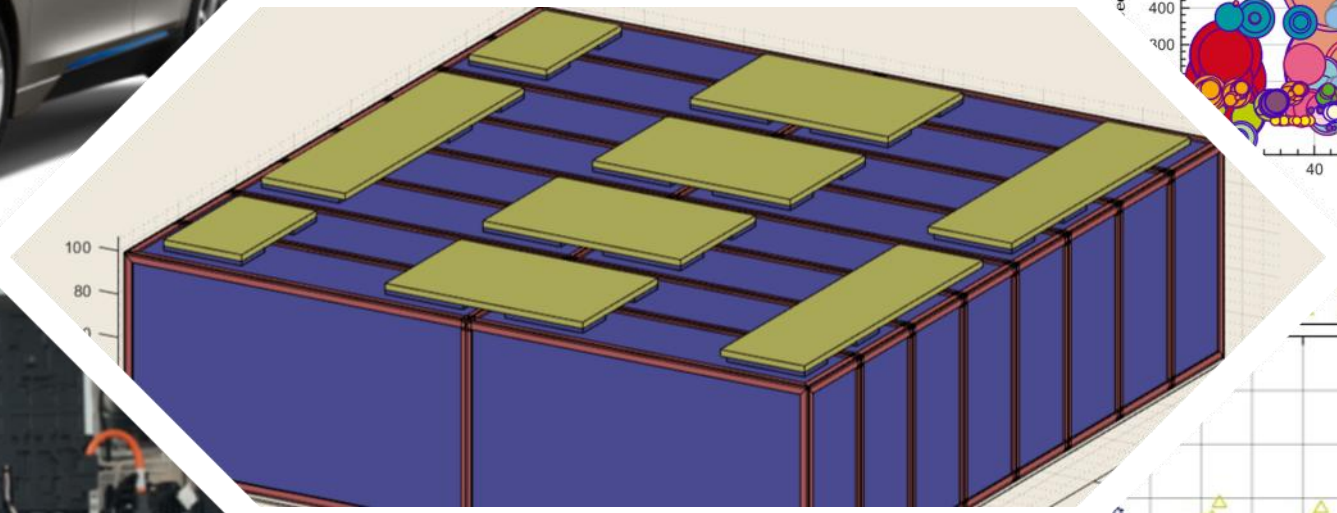


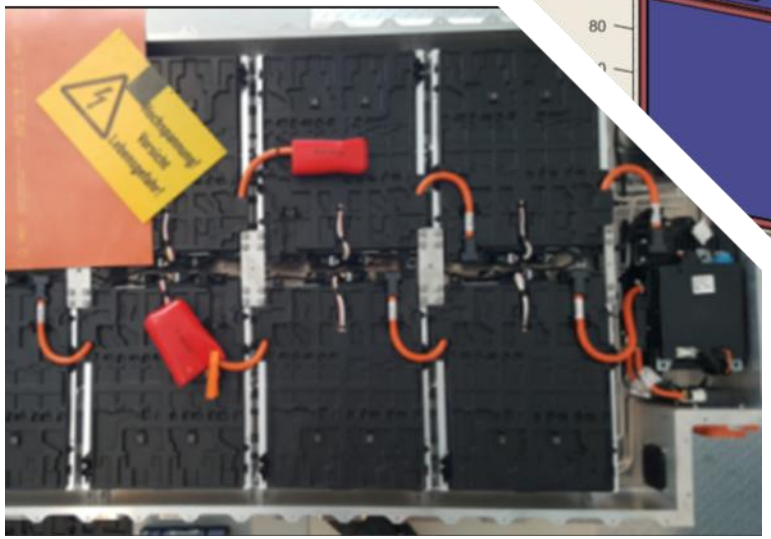
Overview Battery System Design

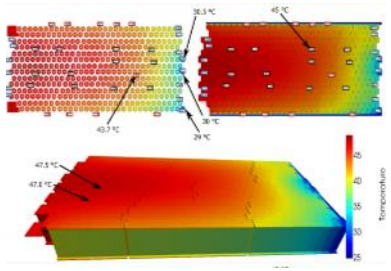


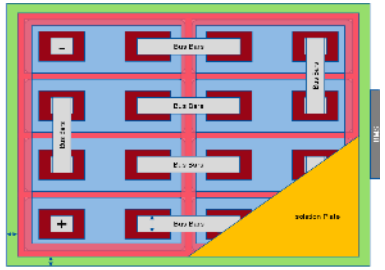


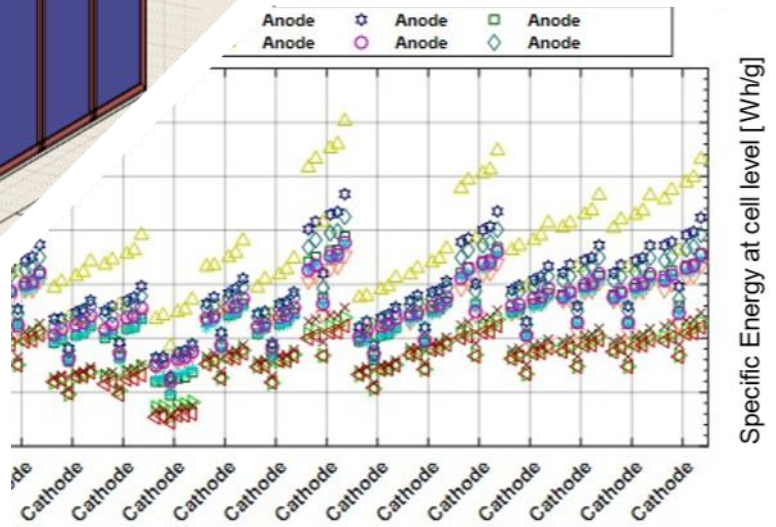








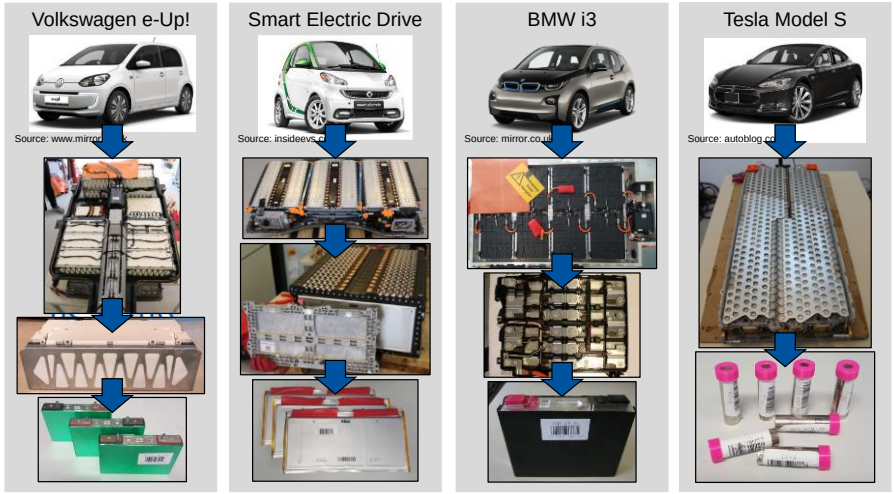




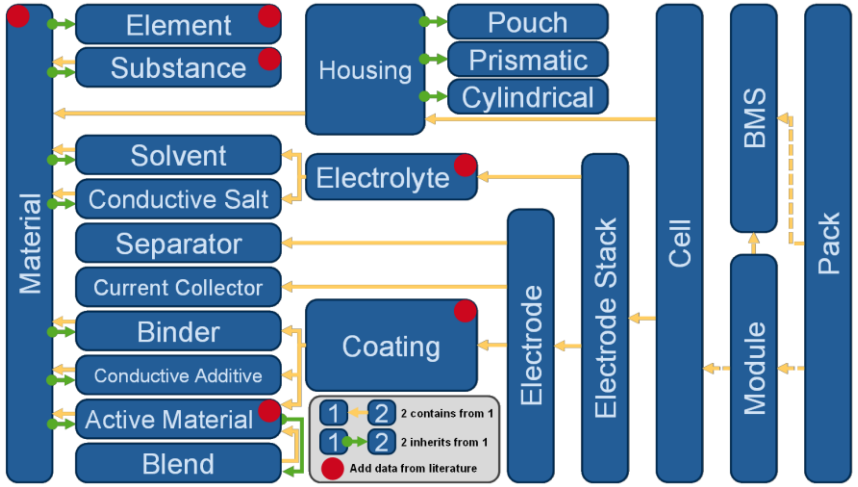
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Battery System Design - Overview

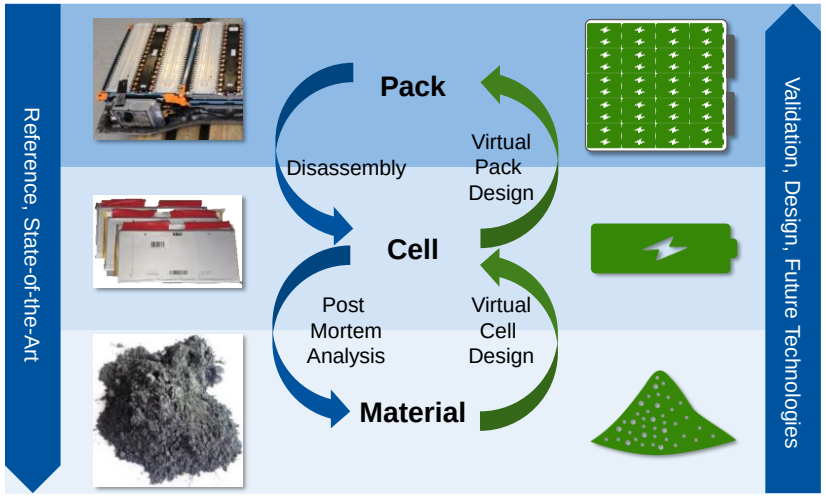
Teardown of Real World Electric Cars



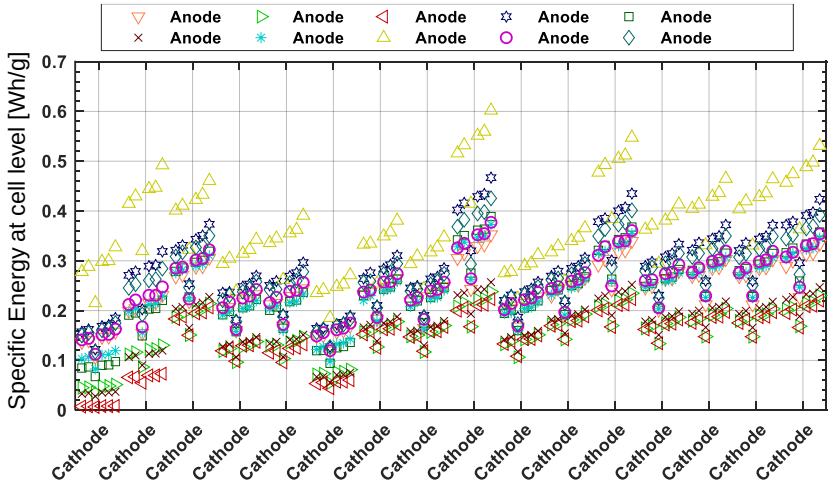
Database to Create Virtual Cells



Digital Twin of Batteries



Benchmarking



Multiple Applications

Battery-electric trains



81 % (regional, Germany)

> 500 kWh

< 5 MW

Battery-electric ships



Low percentage

450 – 7.000 kWh

< 0,9 – 17 MW

Battery-electric construction machines



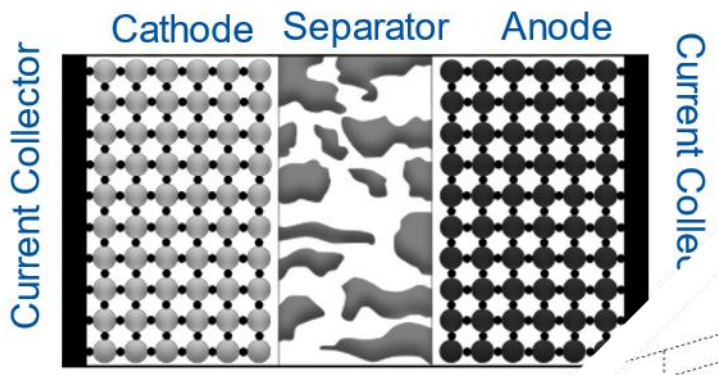
Low percentage

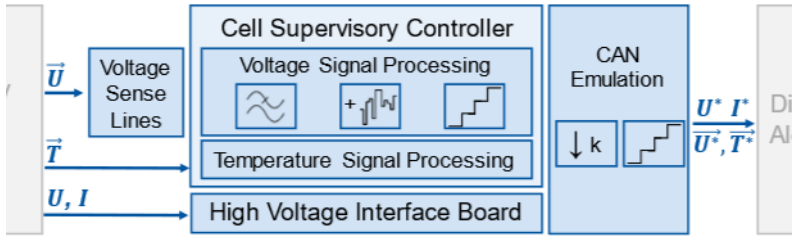
10 – 320 kWh

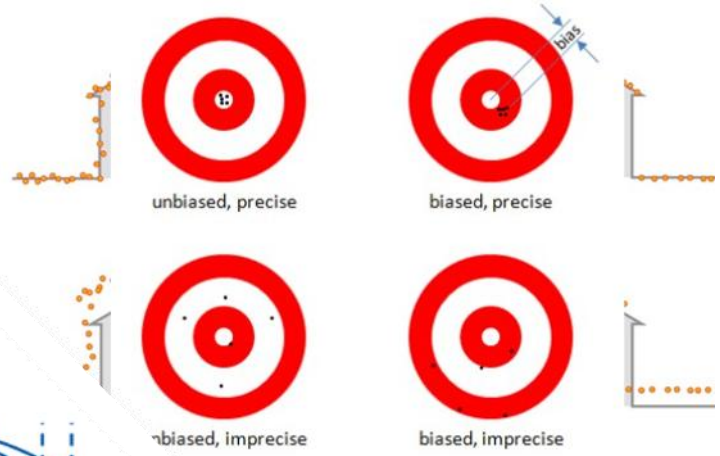
kW to MW

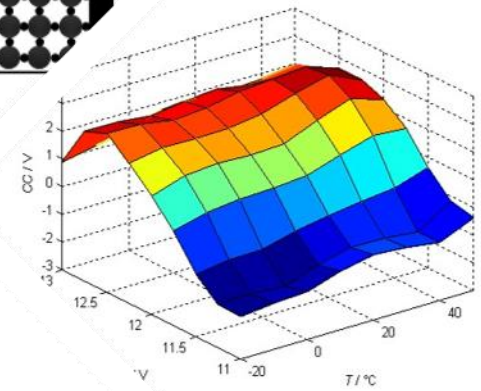
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terex.com/fuchs/
manager-magazin.de/
ingenieur.de/technik/

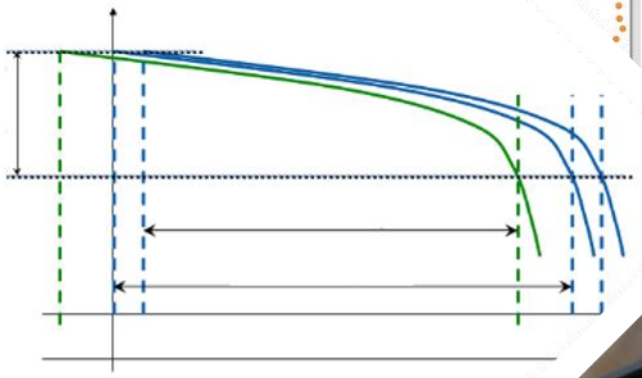
Overview Battery Diagnostics

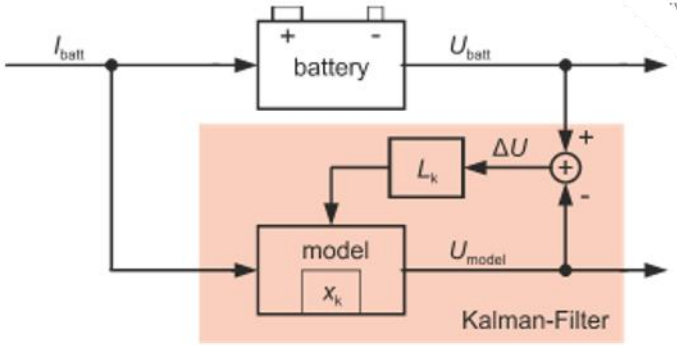


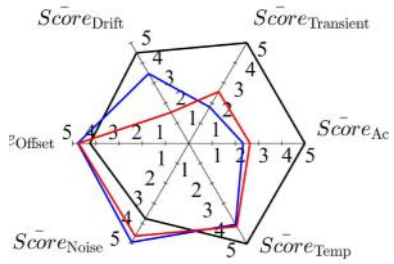


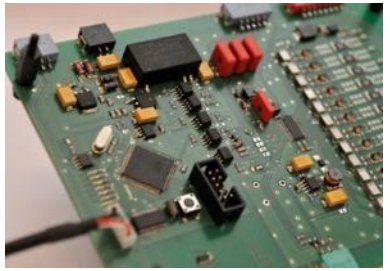


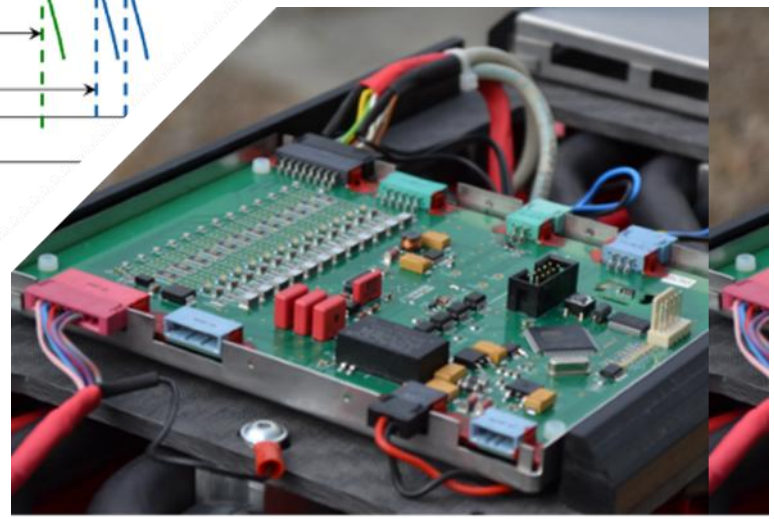






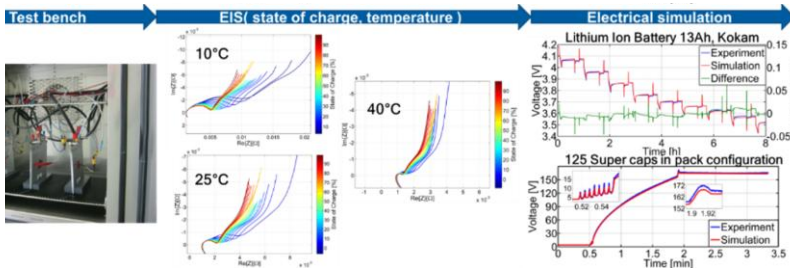




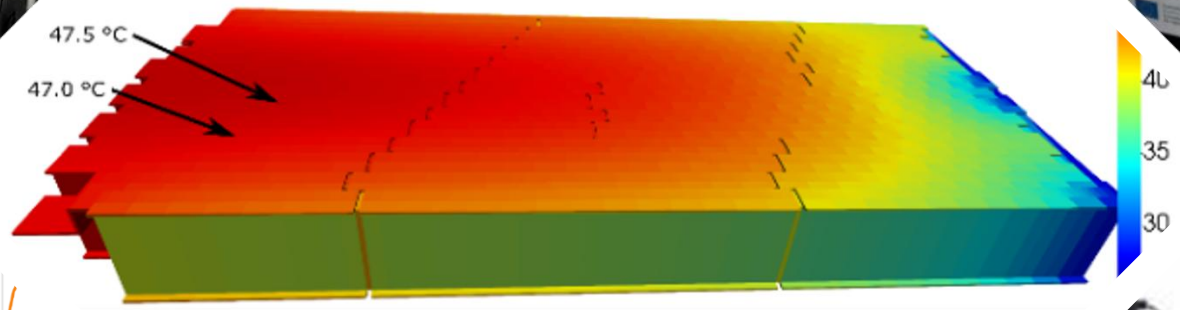


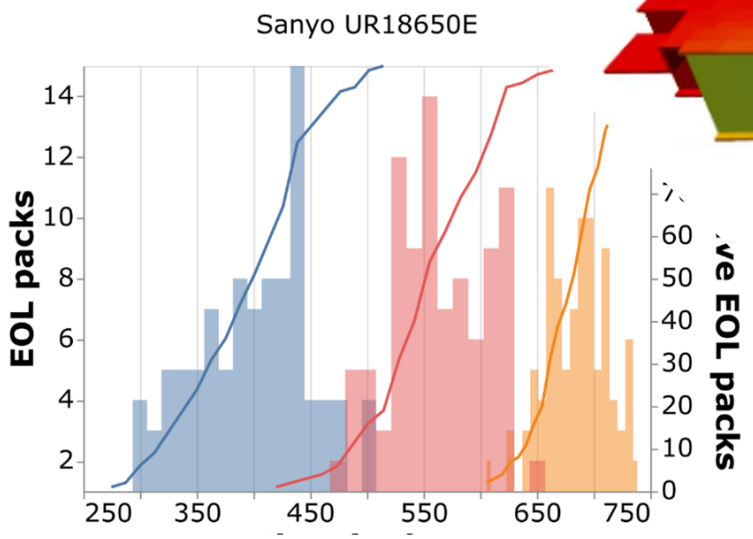
Overview Battery Model Parametrization

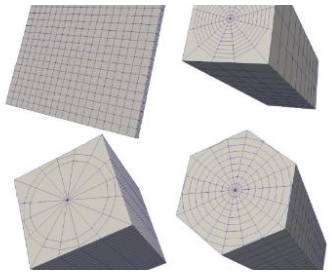












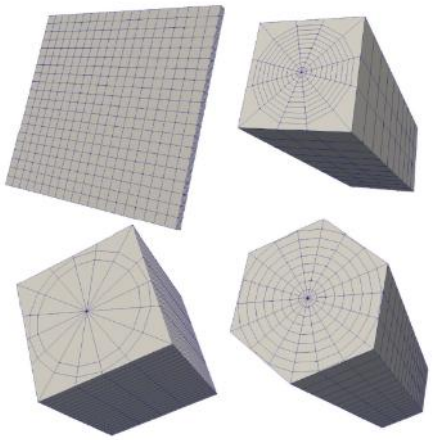




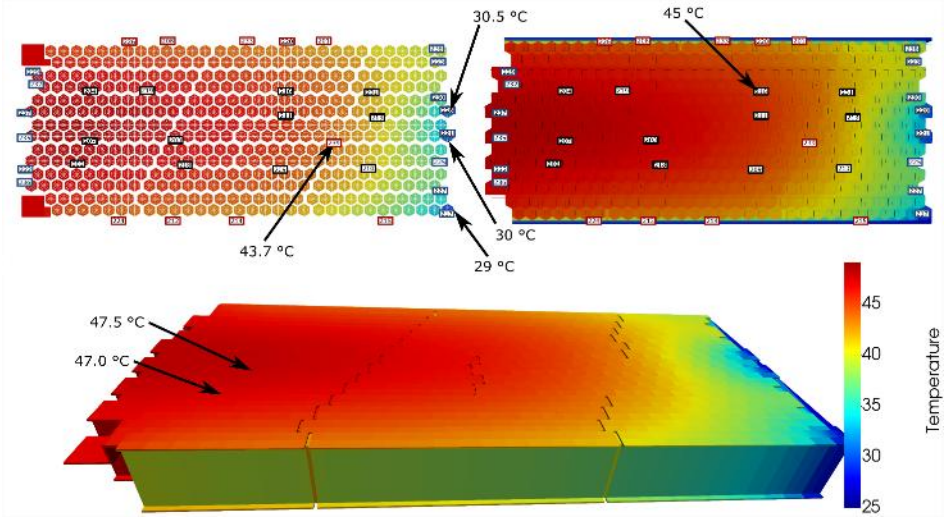
Thermal Model

- Complex geometries can simulated with a composition of simple shapes by finite volumes
- Single Cells can be assembled to packs for simulation
- Thermal parametrization of heat capacity and temperature and heat conductivity with IR-Temperature measurement

Available Volumes



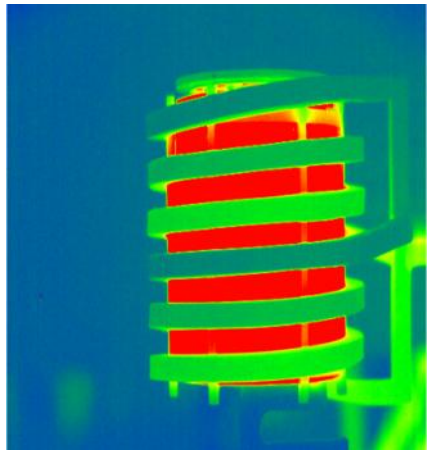
Simulated Pack



Source: Dissertation Hust 2018 Dissertation Witzenhausen 2017

Parametrization





Overview Environmental Load





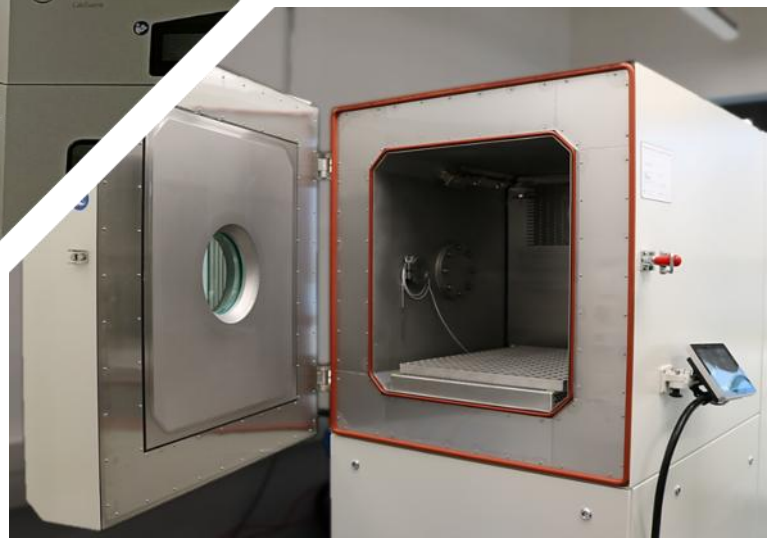




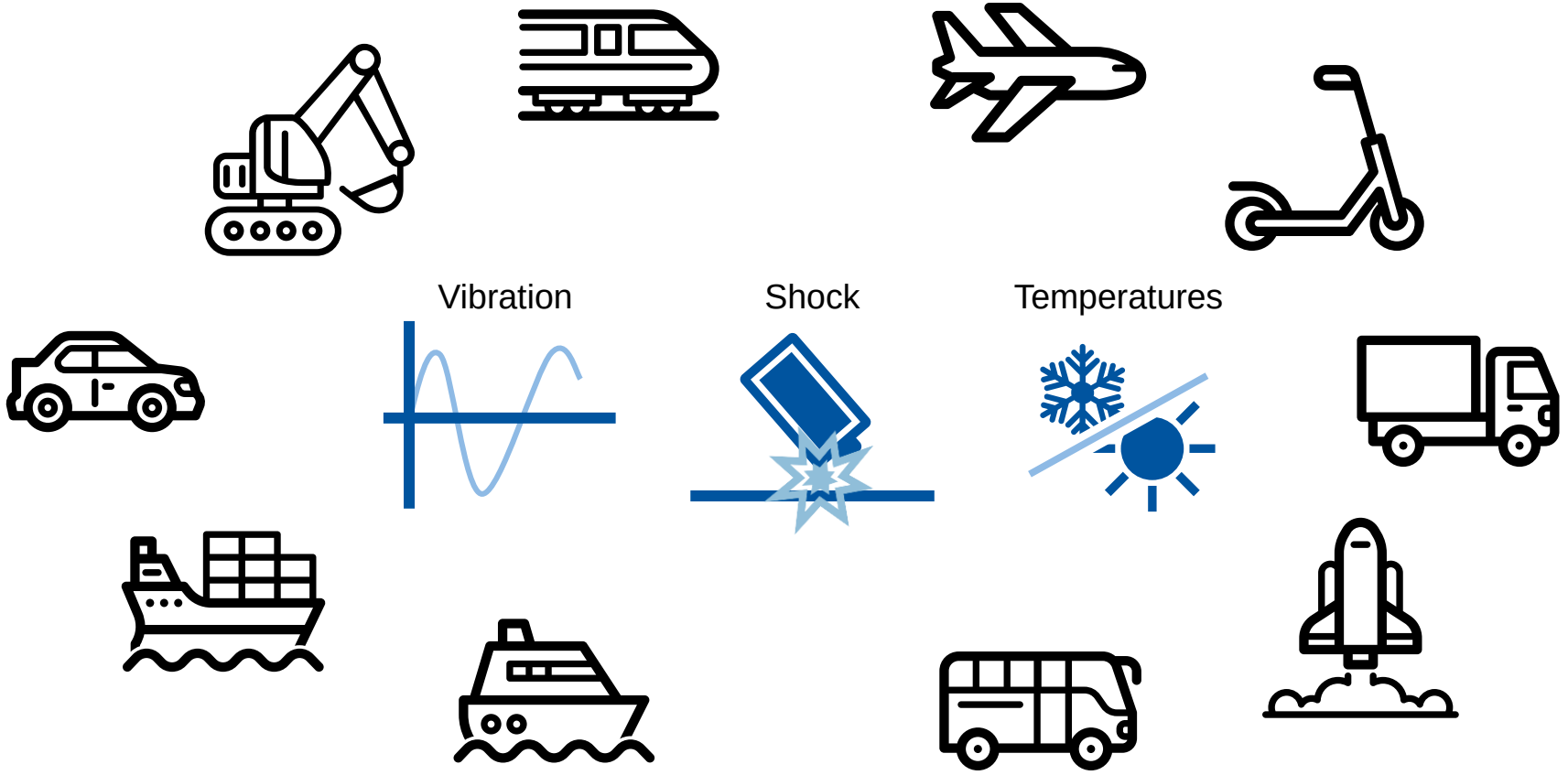








Motivation for mechanical and thermal experiments on batteries



Material

Cell

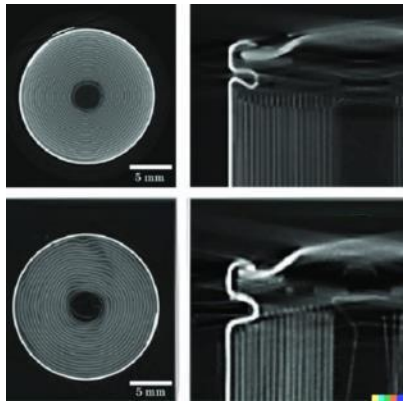
System

Application

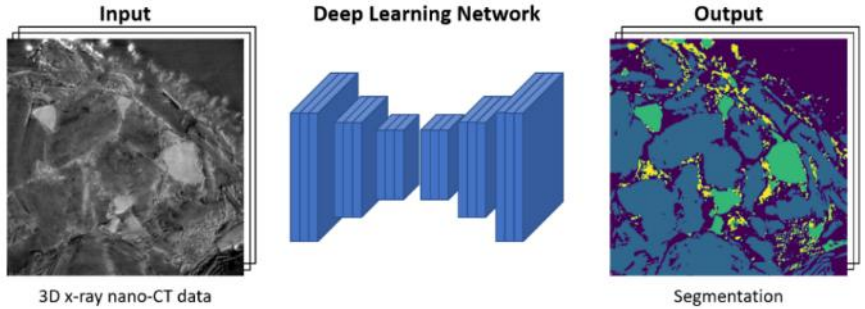
Batteries must withstand difficult environments, especially in mobile applications.

Machine learning in battery research at material & component level

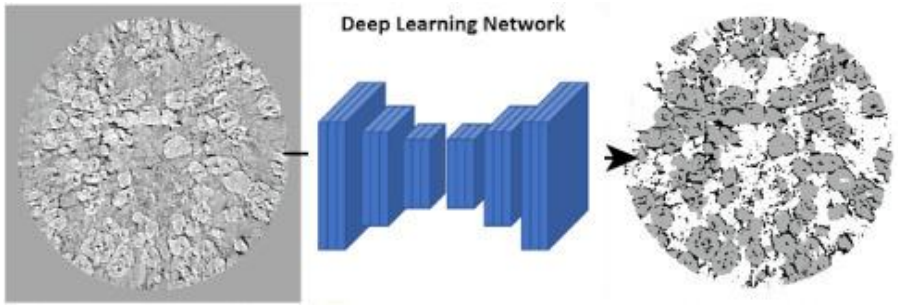
- Anomaly detection in battery production



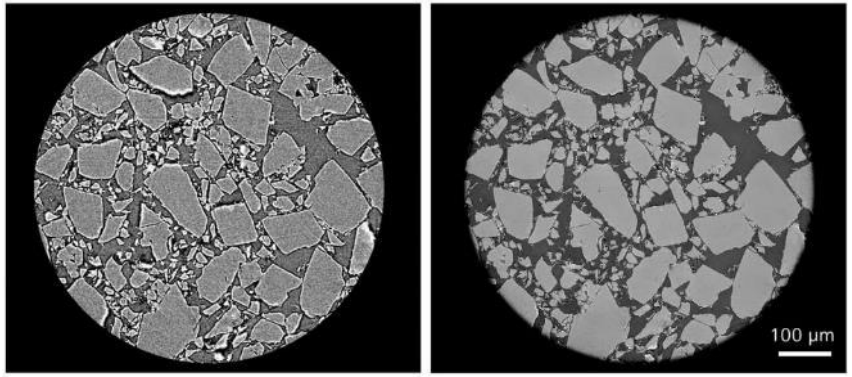
- Supervised segmentation for battery analysis



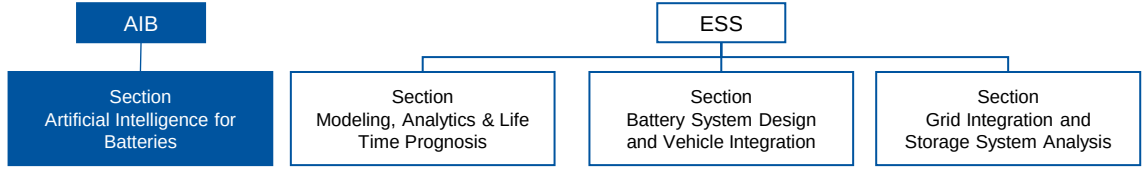
- Denoising CT images with deep learning



- Enhancing CT images with deep learning

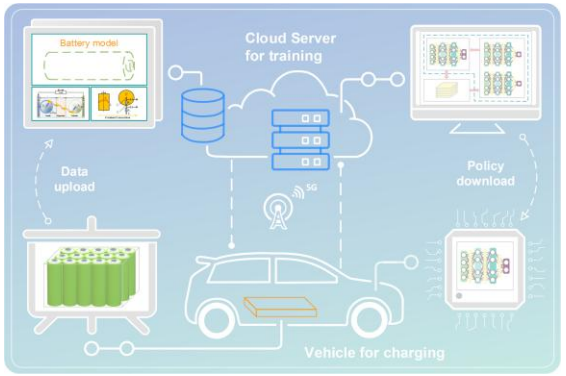


Machine learning in battery research at system level

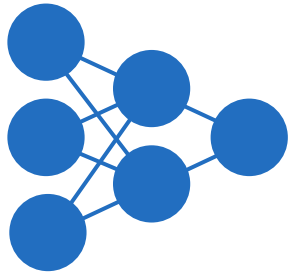


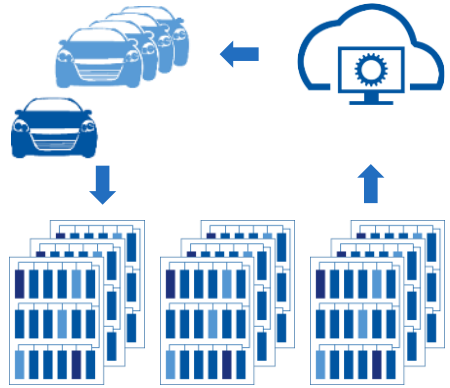
- Fast charging strategy



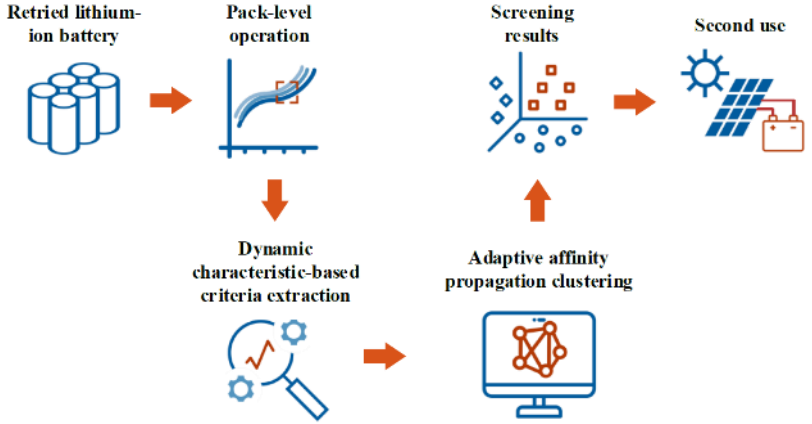


- Operation strategy

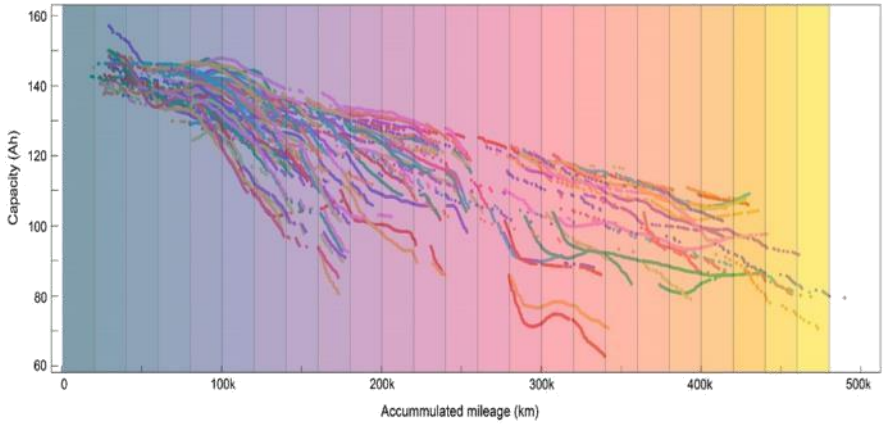




- Fast screening for Second Use



- Field data analysis



Battery Sections at ISEA



Ageing and Lifetime Prediction of Batteries

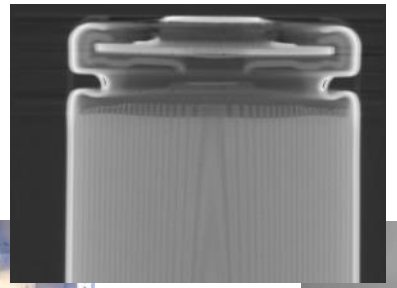
Electrochemical Energy Conversion and Storage Systems (ESS)



Artificial Intelligence and Digitalization for Batteries

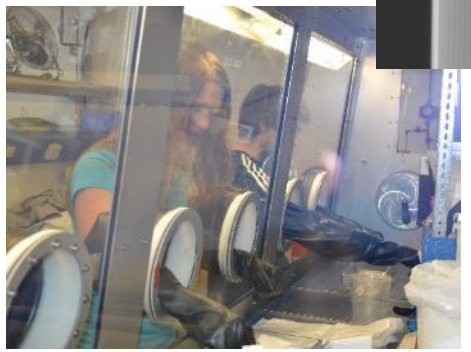


Advanced electrode manufacturing	Ageing and Lifetime Prediction of Batteries	Battery Modelling, Analytics and Lifetime Prediction	Battery System Design and Vehicle Integration	Battery Grid Integration and Storage System Analysis	Artificial Intelligence for Batteries



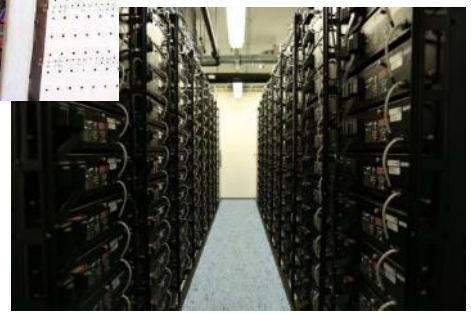






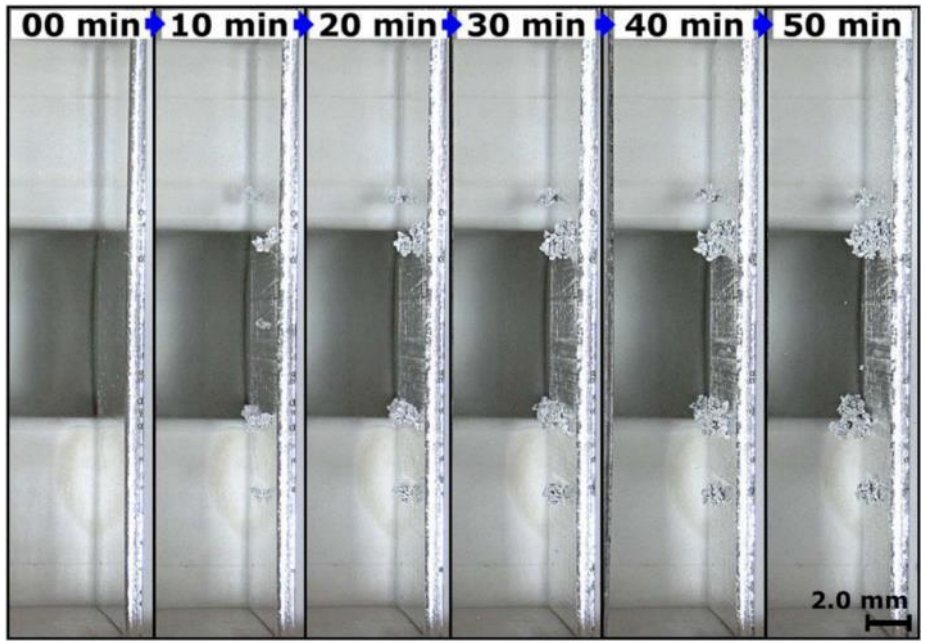




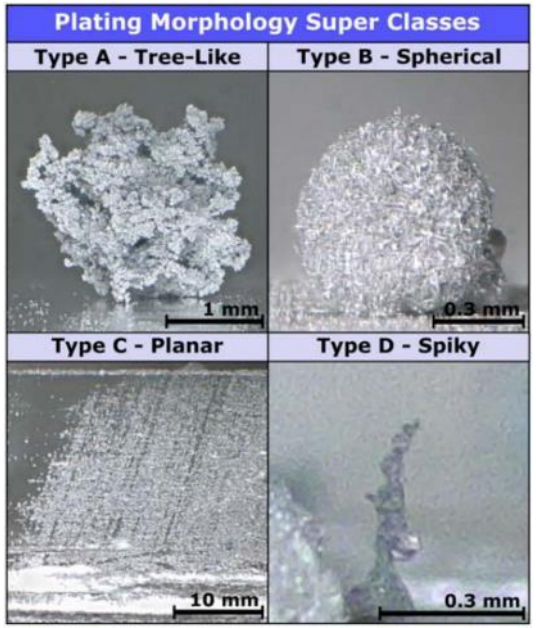


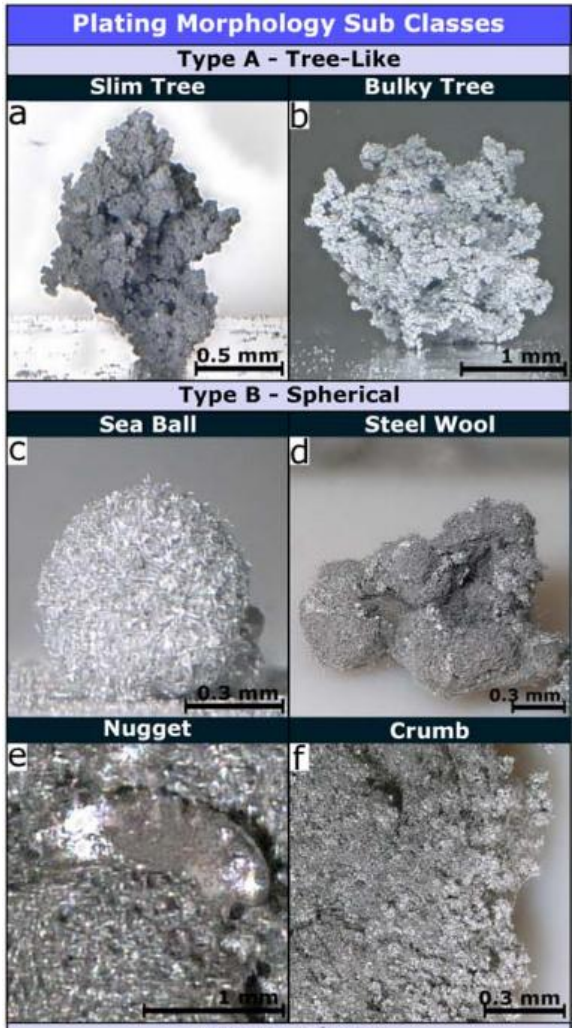
High energy-density anode: Li-metal anode

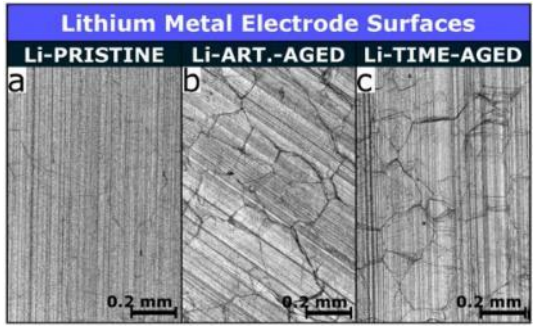
- Dendrite growth
  - Custom designing of the in-situ optical cell
  - Dendrite growth as a function of resting time
  - Understanding and categorization of lithium deposition



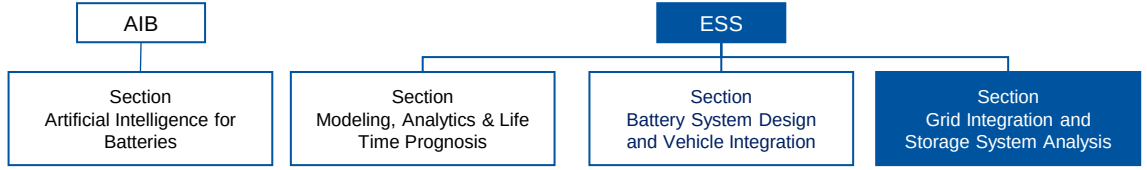
H. Kühnle et al., https://doi.org/10.1149/1945-7111/abdeeb

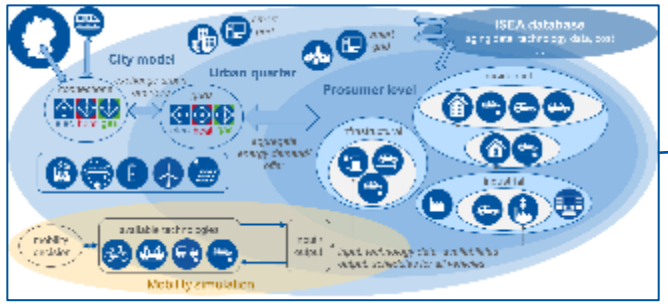




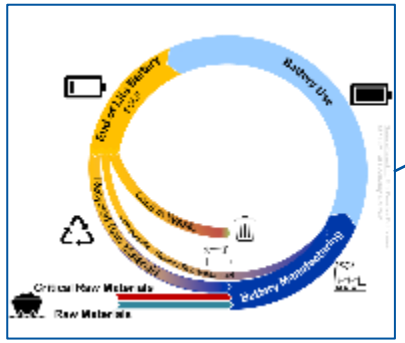


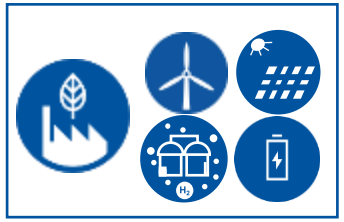
Research at Group Grid Integration and Storage System Analysis











Energy System Modelling

Battery Storage at Utility Scale

Charging Infrastrucuture

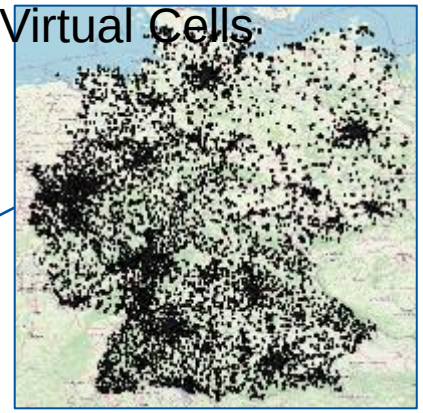
Circular Economy & LCA

PV Storage

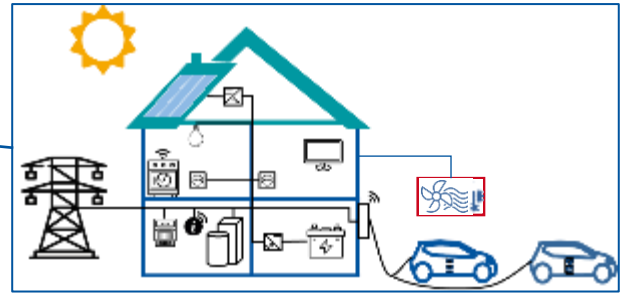
Green Industry

Smart Energy Home

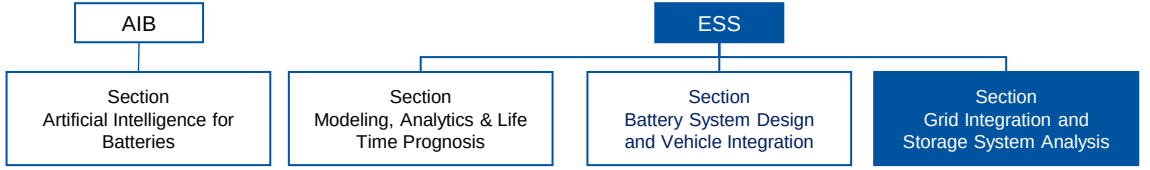
Database to Create Virtual Cells







Charging Infrastructure Analysis

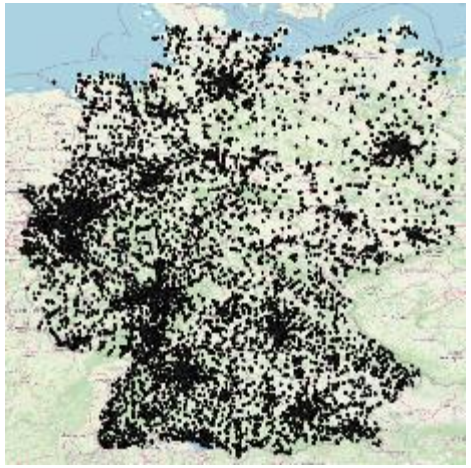


The research field focuses on understanding, analyzing, predicting and influencing the use of charging infrastructure.

Public Charging Infrastructure

- BeNutz LaSA (BMWK)
  - Analysis of the majority of the public charging infrastructure in Germany with a view to usage patterns, energy requirements and optimization opportunities.
  - Methods from the fields of Big Data and machine learning
  - Forecasting the probability of occupancy
  - Forecast of good new locations
  - Partner: FCN-ECO, IAEW, smart/lab, Hubject

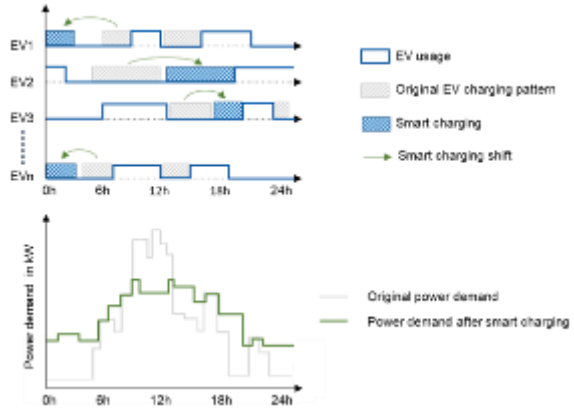
Funding: Bundesministerium für Wirtschaft und Klimaschutz



Fleet Charging Infrastructure

- FlexFleet (BMDV)
  - Analysis of the flexibilization potential of commercial vehicle fleets based on more than 100 vehicle fleets
  - Field test with intelligent charging algorithms as well as flexible network elements
  - Partner: BIT in Peking

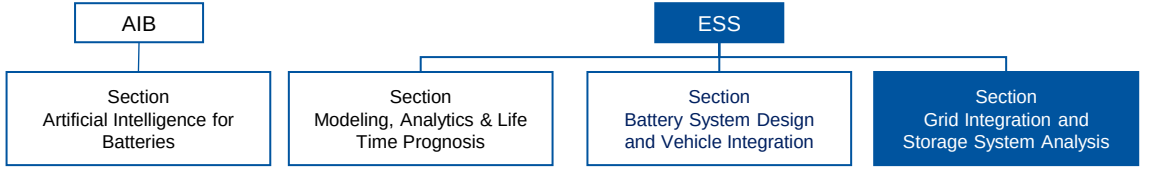
Funding: Bundesministerium für Digitales und Verkehr; Bundesministerium für Wirtschaft und Klimaschutz

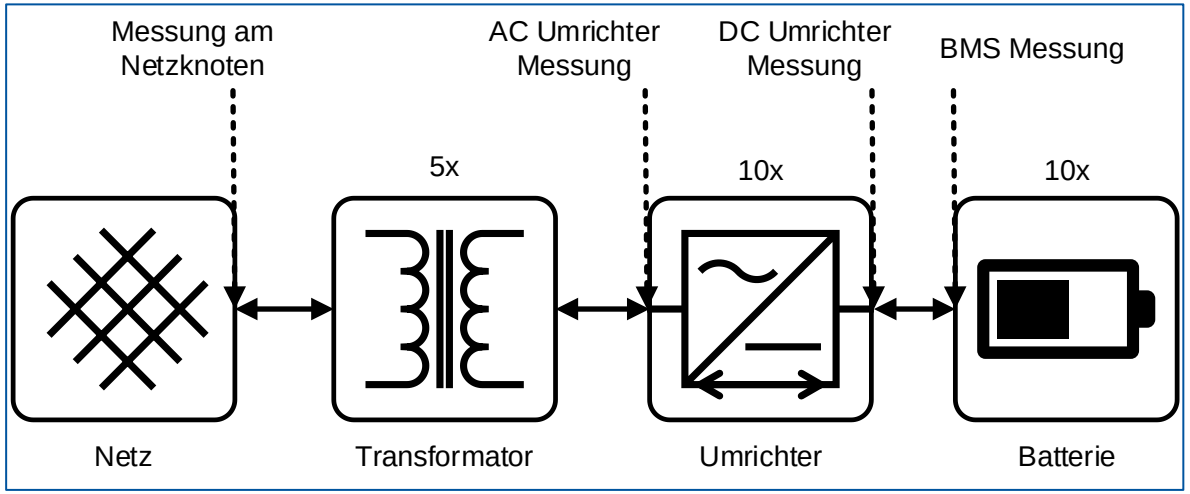


- Finalize (BMWK)
  - Conversion of an airport fleet to electric vehicles
  - Testing of the flexibility potential as well as consideration of the LCA of the vehicles with different charging strategies.

Utility Scale Research Battery Storage – M5Bat







Technical data

- Power: approx. 6 MW
- Energy: approx. 7.5 MWh
- 5 Battery technologies (Li-Ion & lead-acid)
- 10 independent battery units

Operation

- System services
- 3 MW FCR (frequency containment reserve)
- Intraday trading (15 min market)
- In preparation: aFRR (frequency restoration)

Overview of sector coupling in industry applications: everything is connected!





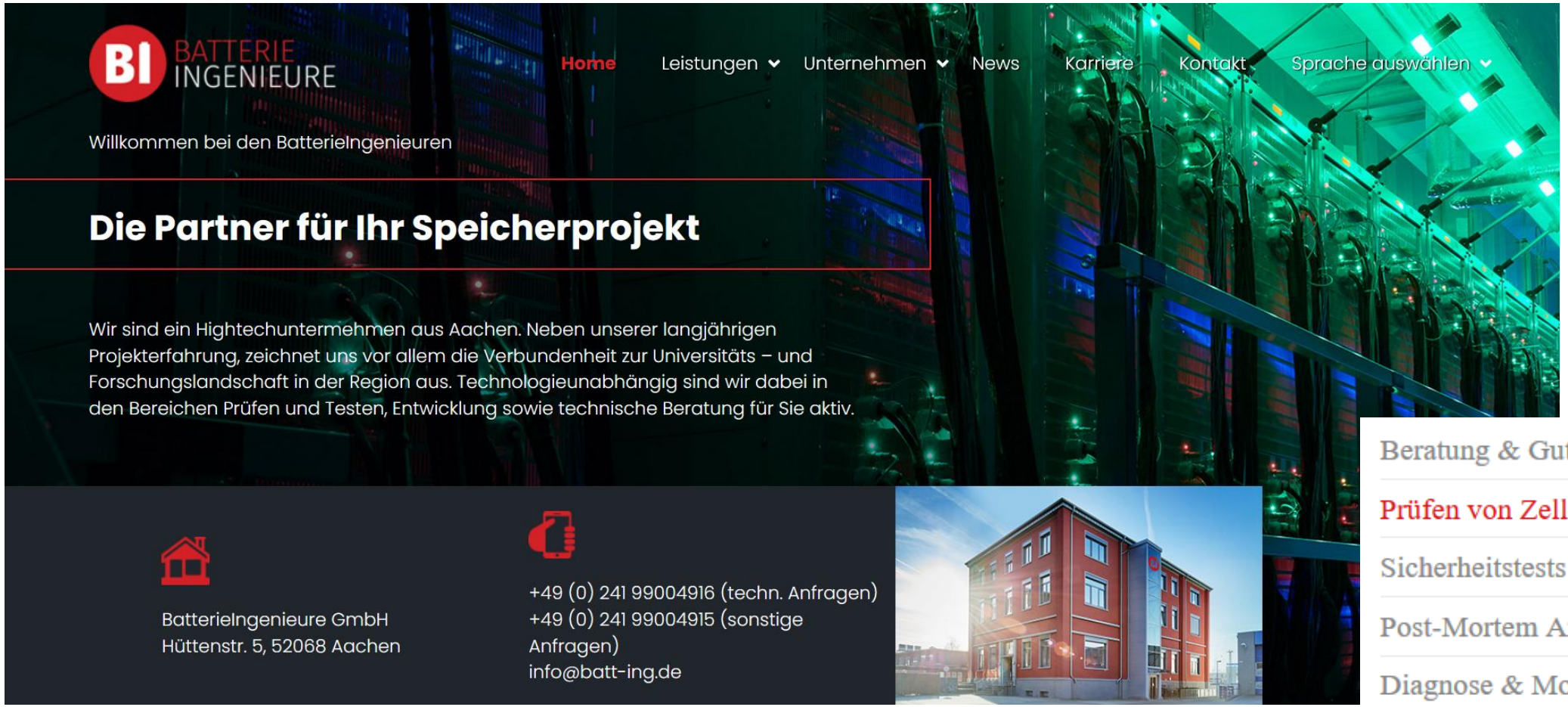




Co-operation with industry partners in many different industrial areas (incomplete)

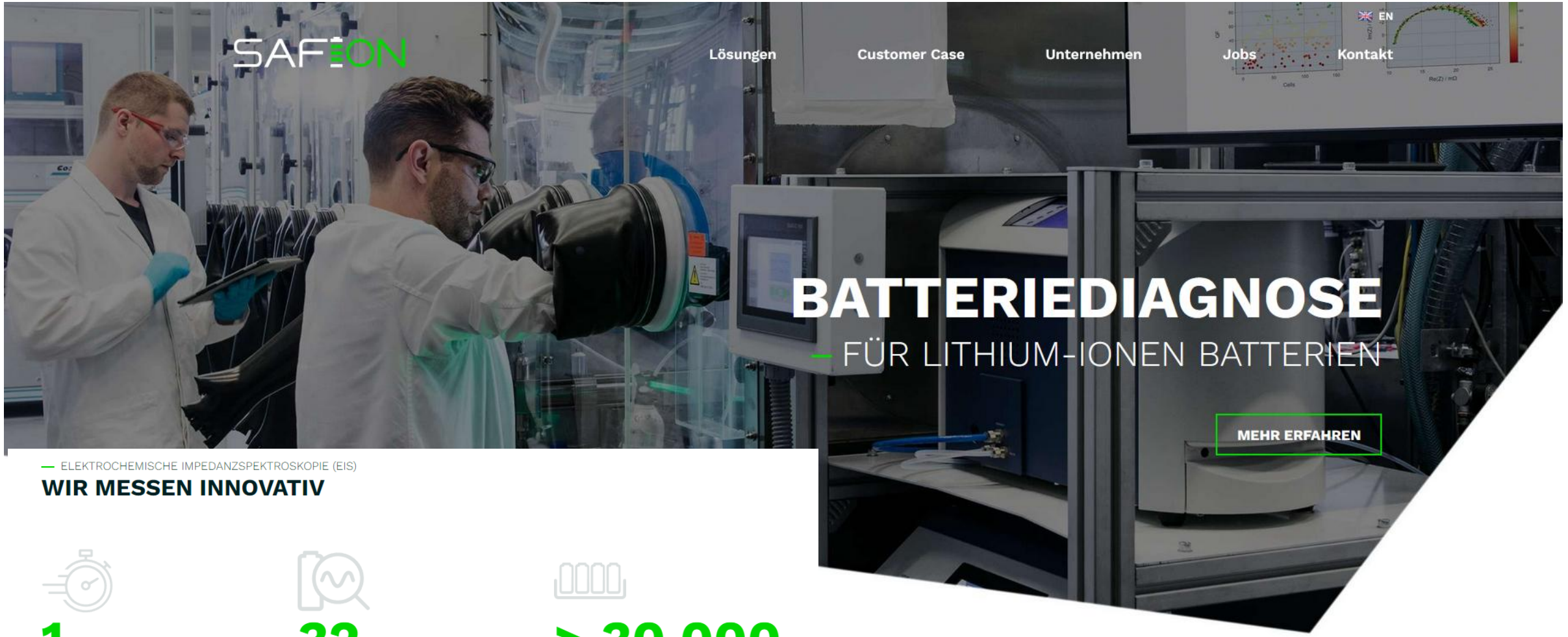


Spin-off “Batterieingenieure GmbH” – Services in Industriestandards



- Beratung & Gutachten
- Prüfen von Zellen und Batterien
- Sicherheitstests
- Post-Mortem Analyse
- Diagnose & Modellierung
- Qualitätssicherung
- Batterieentwicklung und -fertigung

Spin-off “Safion GmbH” – Zustandsbestimmung & Qualitätskontrolle



Spin-off “ebusplan GmbH” – Einsatz- und Umsatzplanung für E-Busse



Spin-off “ACCURE GmbH” – Datenanalyse mit Machine Learning

ACCURE BATTERY INTELLIGENCE

Empower your batteries

The industry of the 21st century is built on batteries. To understand and improve the safety, reliability and lifetime of batteries, companies around the world rely on ACCURE's actionable data analytics.

LEARN MORE



Masterstudiengang „Battery Science and Technology in Engineering“

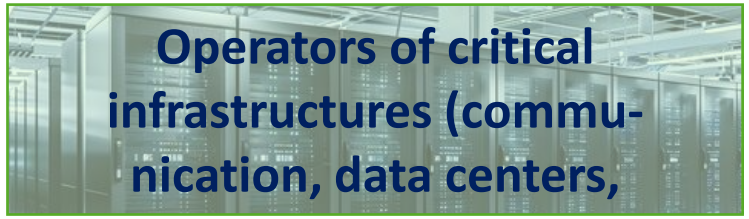




























Battery development & teaching are highly interdisciplinary

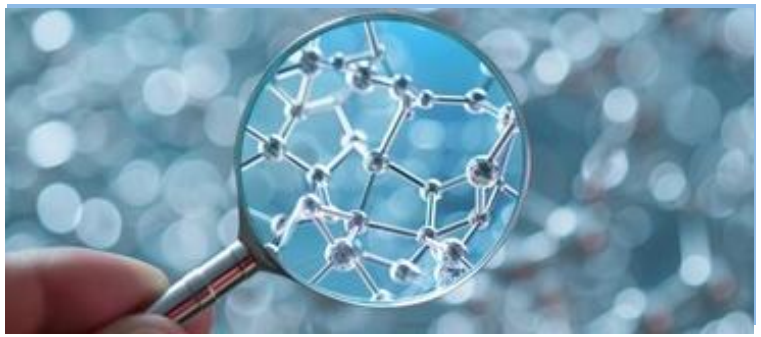












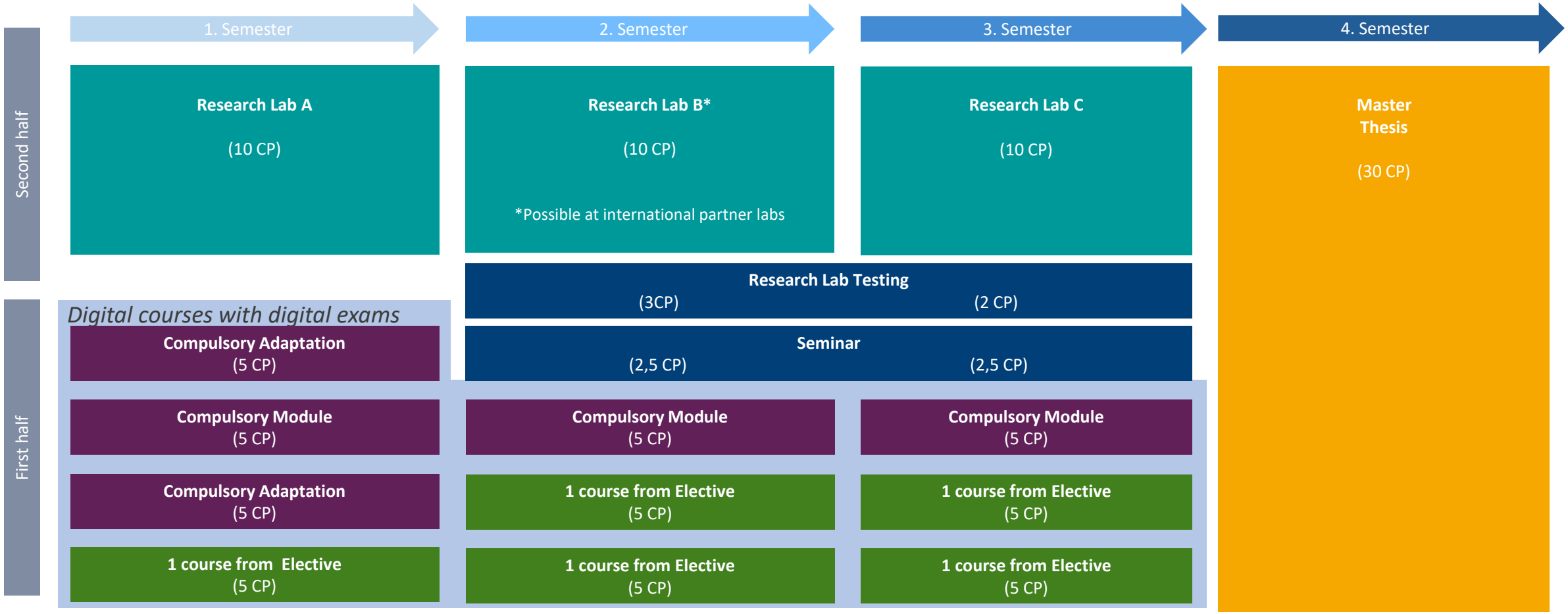






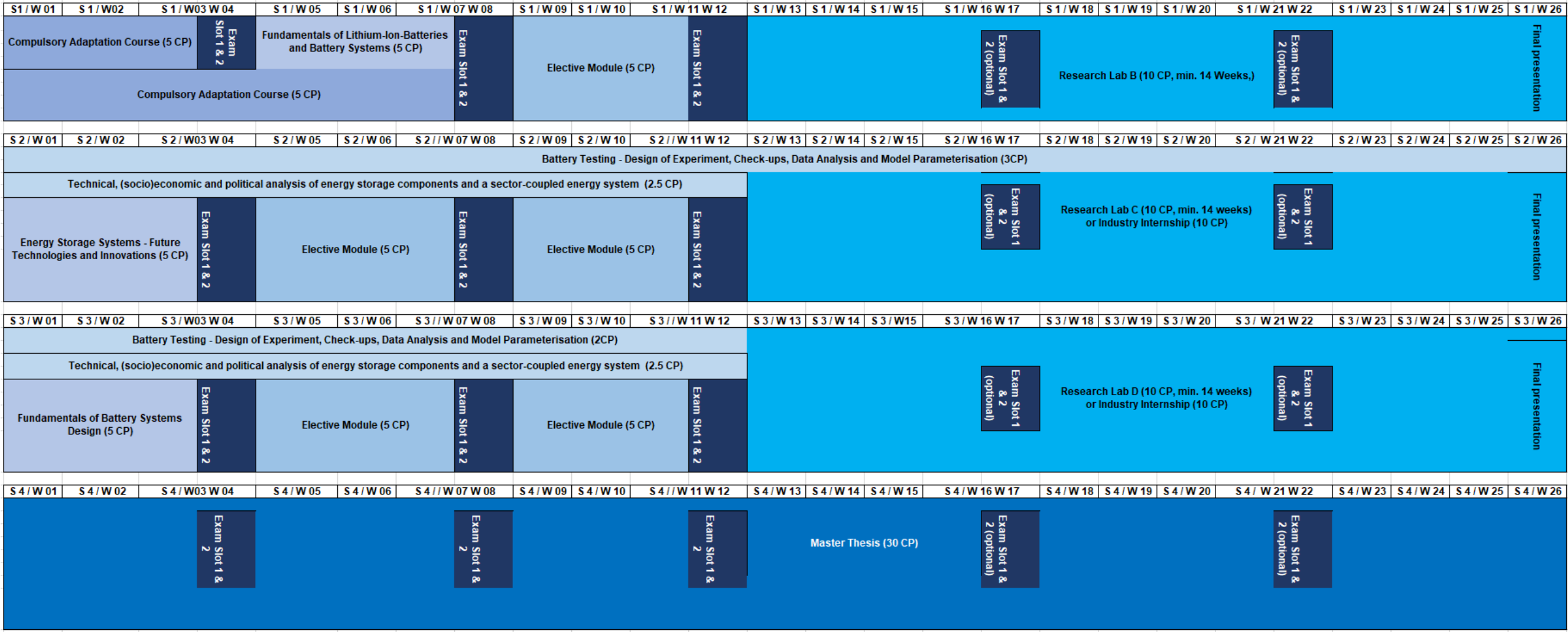
BSTE

Curriculum „Battery Science and Technology in Engineering“



ISEA Stromrichtertechnik und Elektrische Antriebe | RWTH AACHEN UNIVERSITY

Curriculum in Block courses



Noch Fragen?
Sie erreichen uns wie folgt:

Kontakt

Prof. Dirk Uwe Sauer
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Fragen, Anregungen, Fehler

Bitte schauen Sie zunächst in unserem Vorlesungs-FAQ im *moodle*-Lernraum, ob es zu Ihrem Anliegen dort schon einen Eintrag gibt.

Wenn das nicht der Fall ist, dann schreiben Sie bitte eine Email an

et1@isea.rwth-aachen.de

Die Antwort wird in unser Vorlesungs-FAQ eingearbeitet und wird damit auch allen anderen zur Verfügung stehen. Das Vorlesungs-FAQ wird also kontinuierlich wachsen.

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